

Region Convolution Neural Network

Faster R-CNN Mask R-CNN

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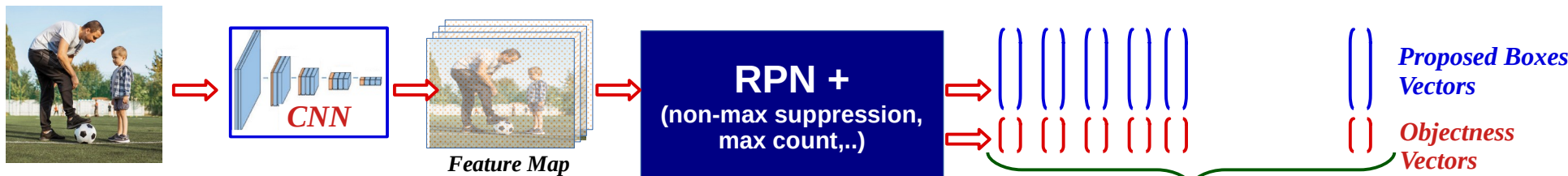


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University of Science and Technology

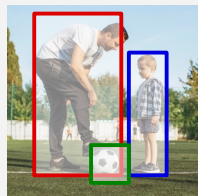
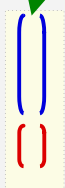
Training of Faster RCNN Detector (A Review)



Run RPN on a Randomly-selected training Image
Filter overlapped Proposed Regions using Non-Max-Suppression
Limit the number of proposed regions to a predefined number

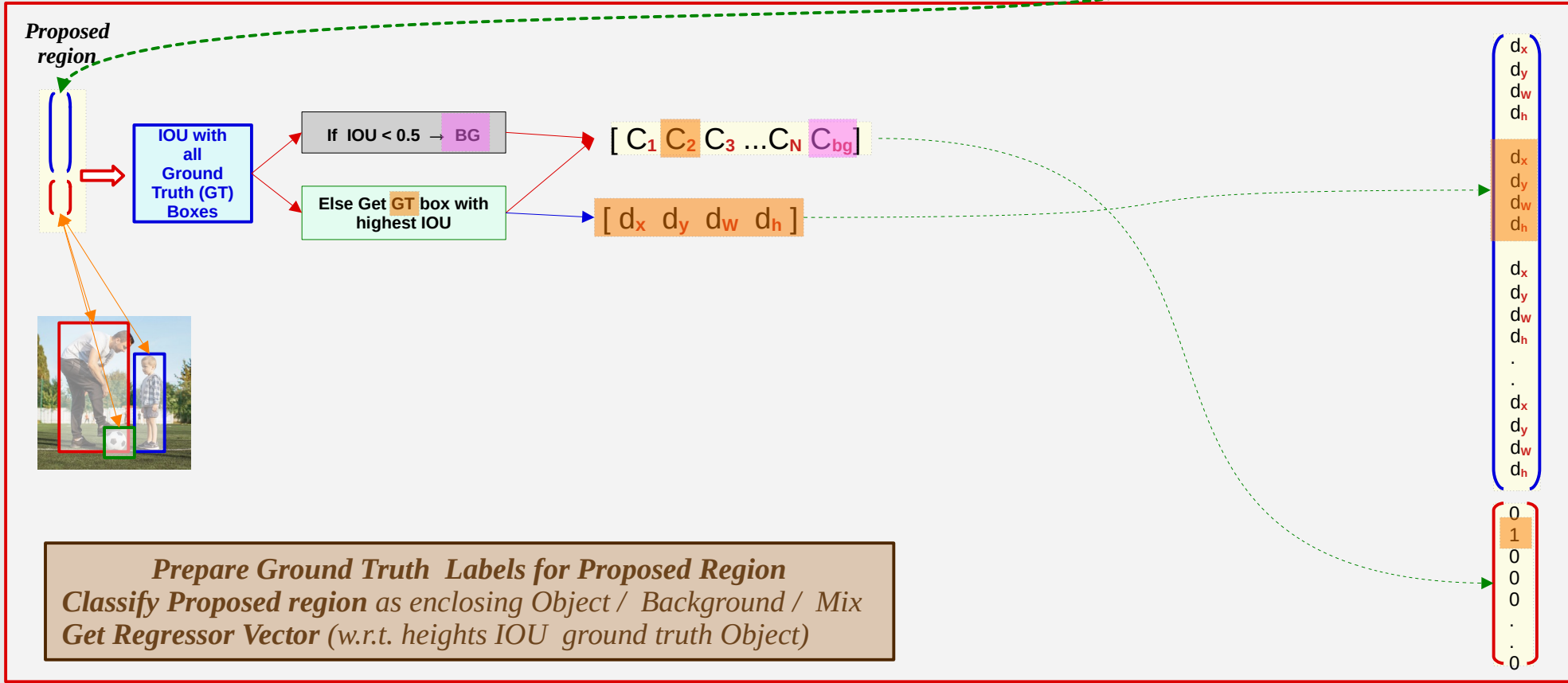
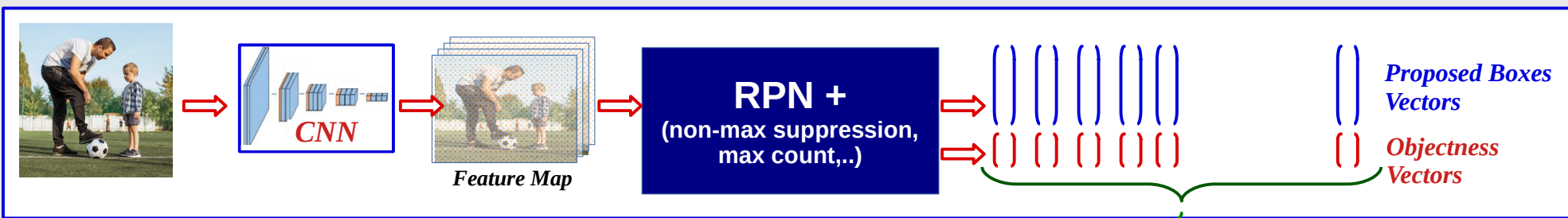


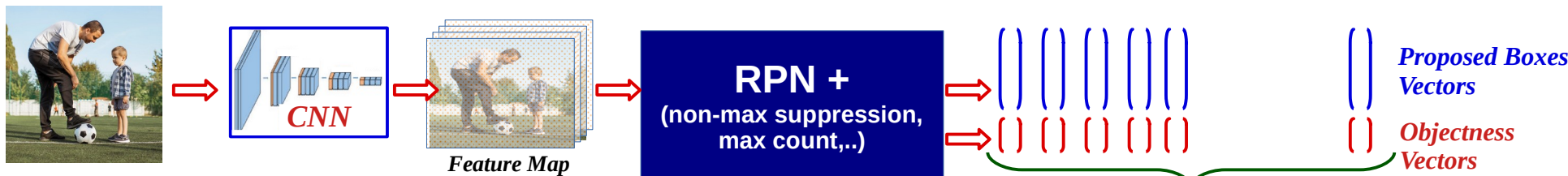
Proposed region



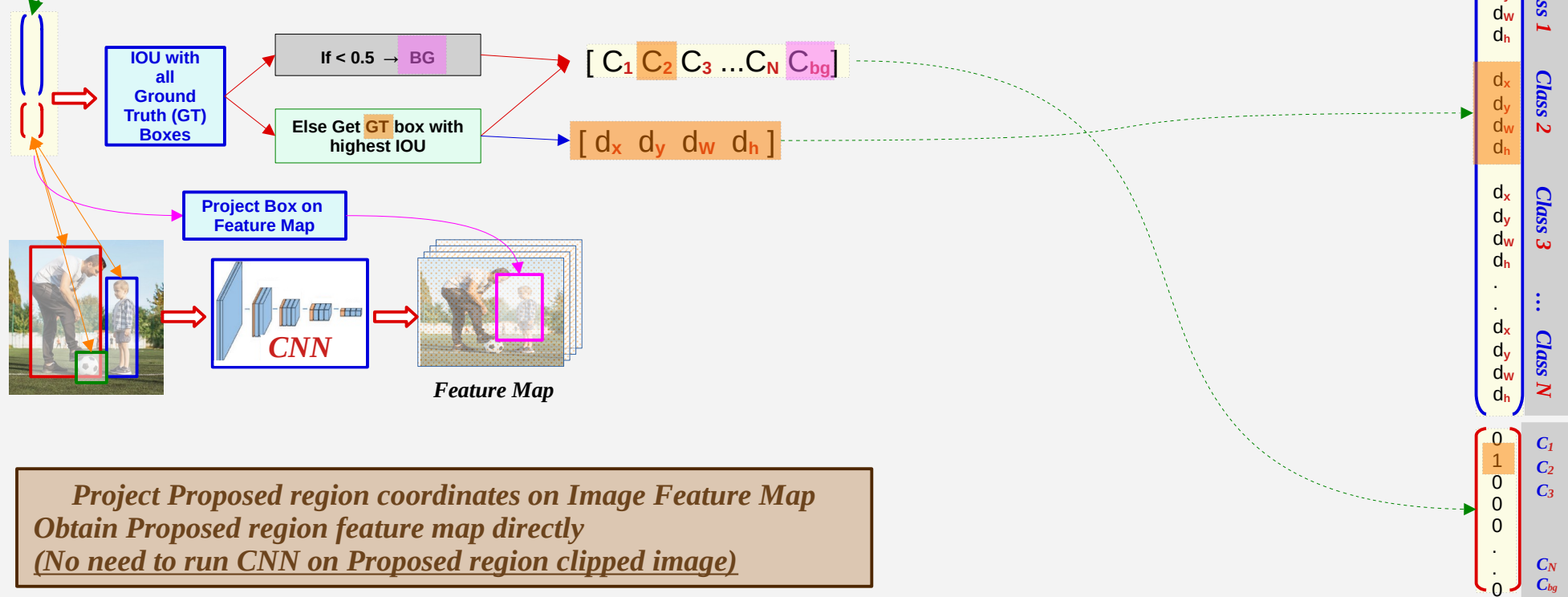
Pick Proposed Region Vectors (one by one)

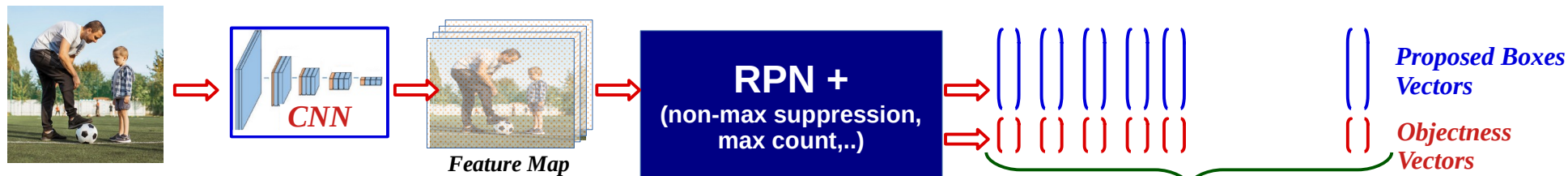
*Supply Proposed region Vectors (**Regressor and Objectness**) in conjunction with **Original Image** to Train Fast RCNN Detector*



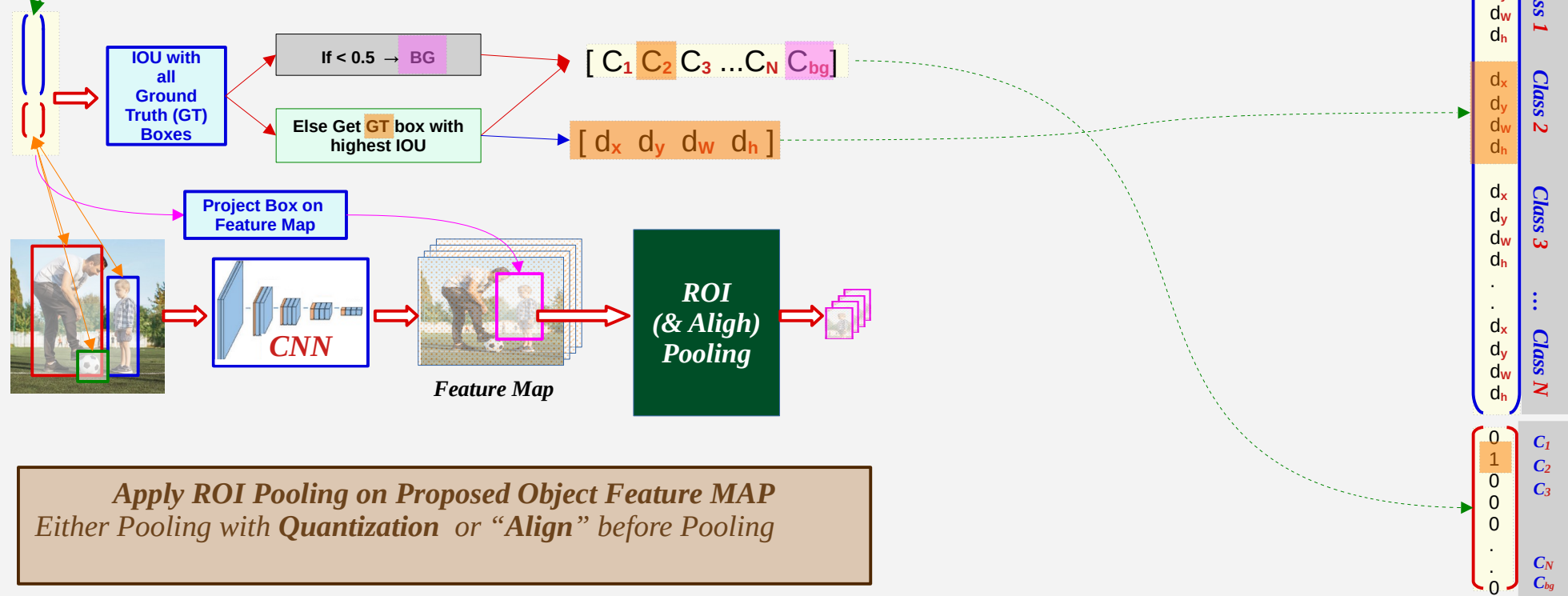


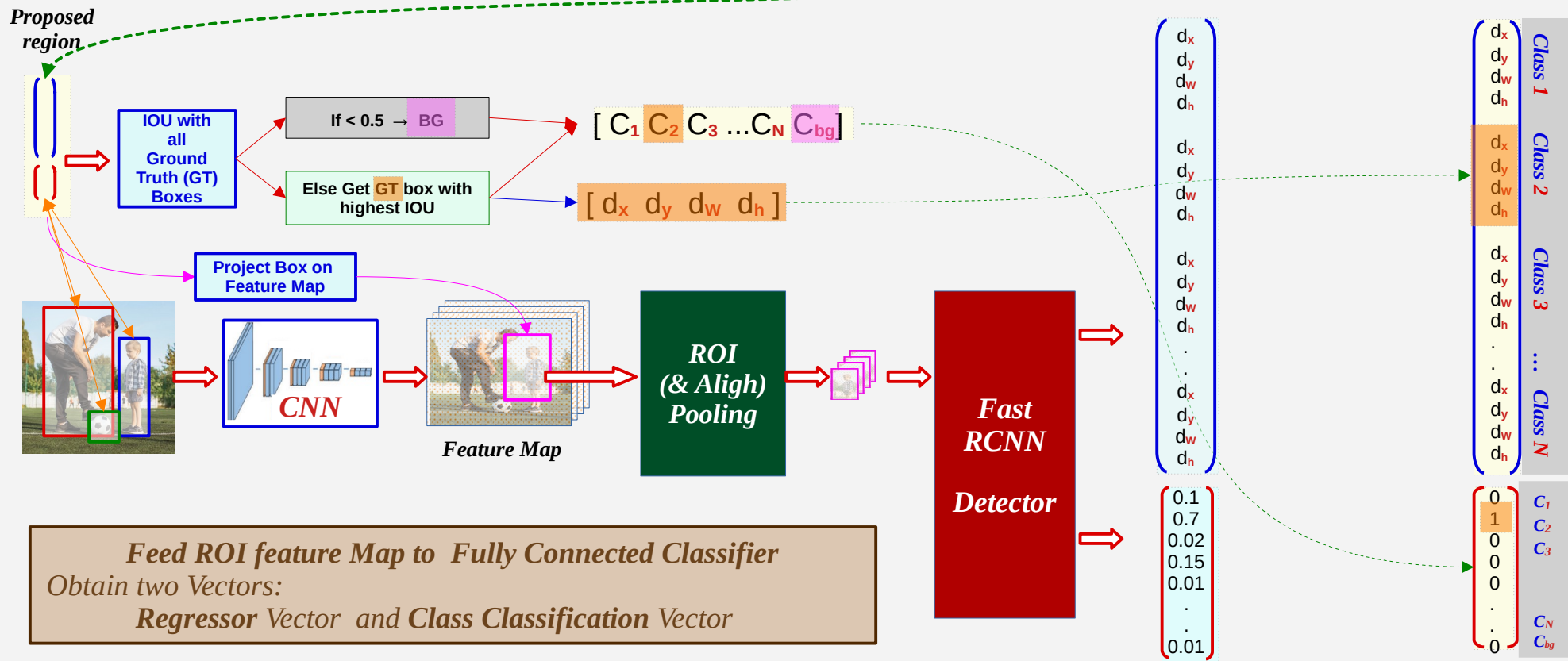
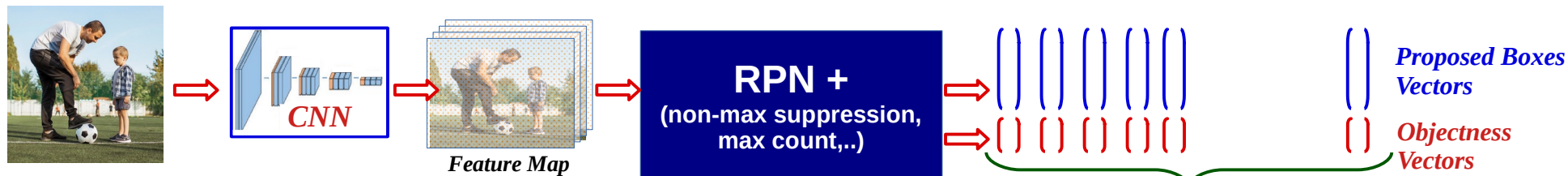
Proposed region

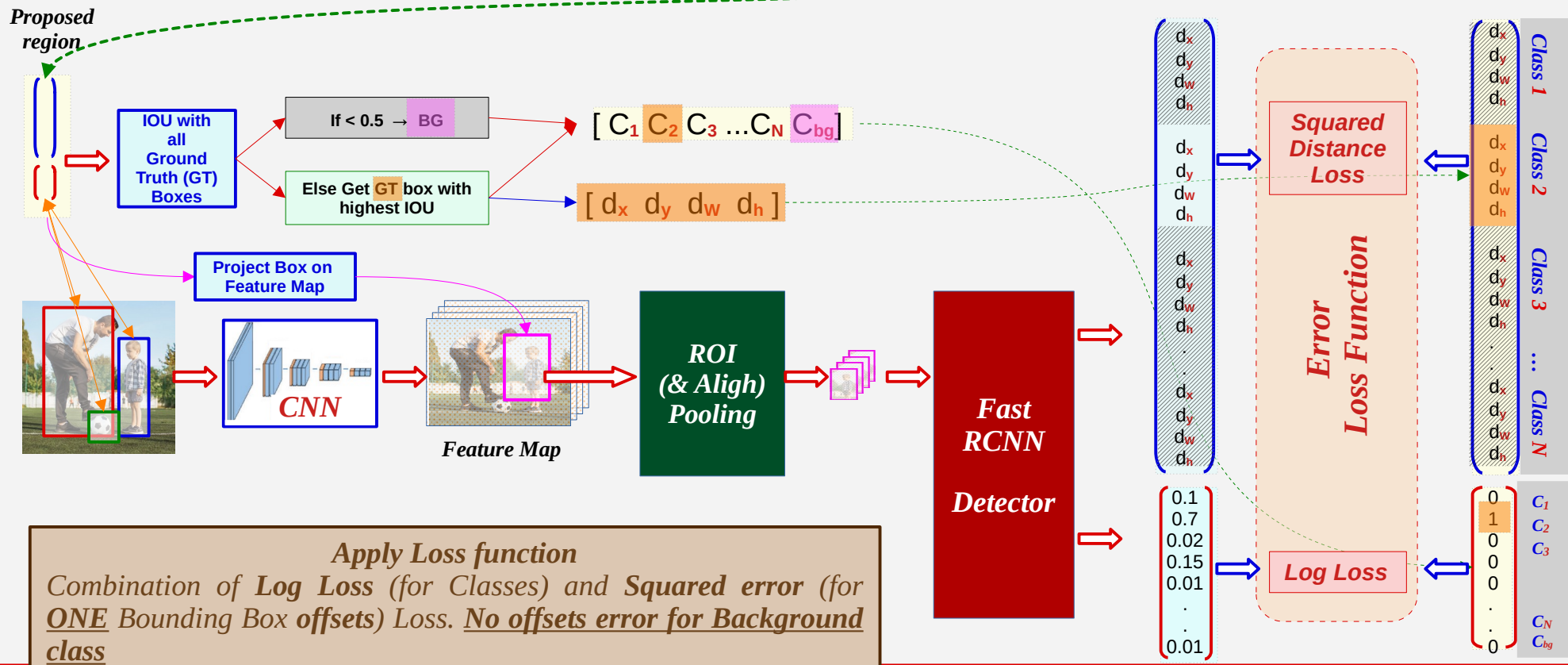
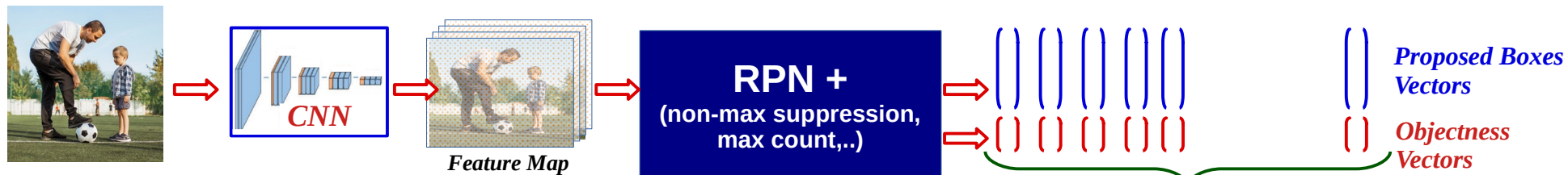


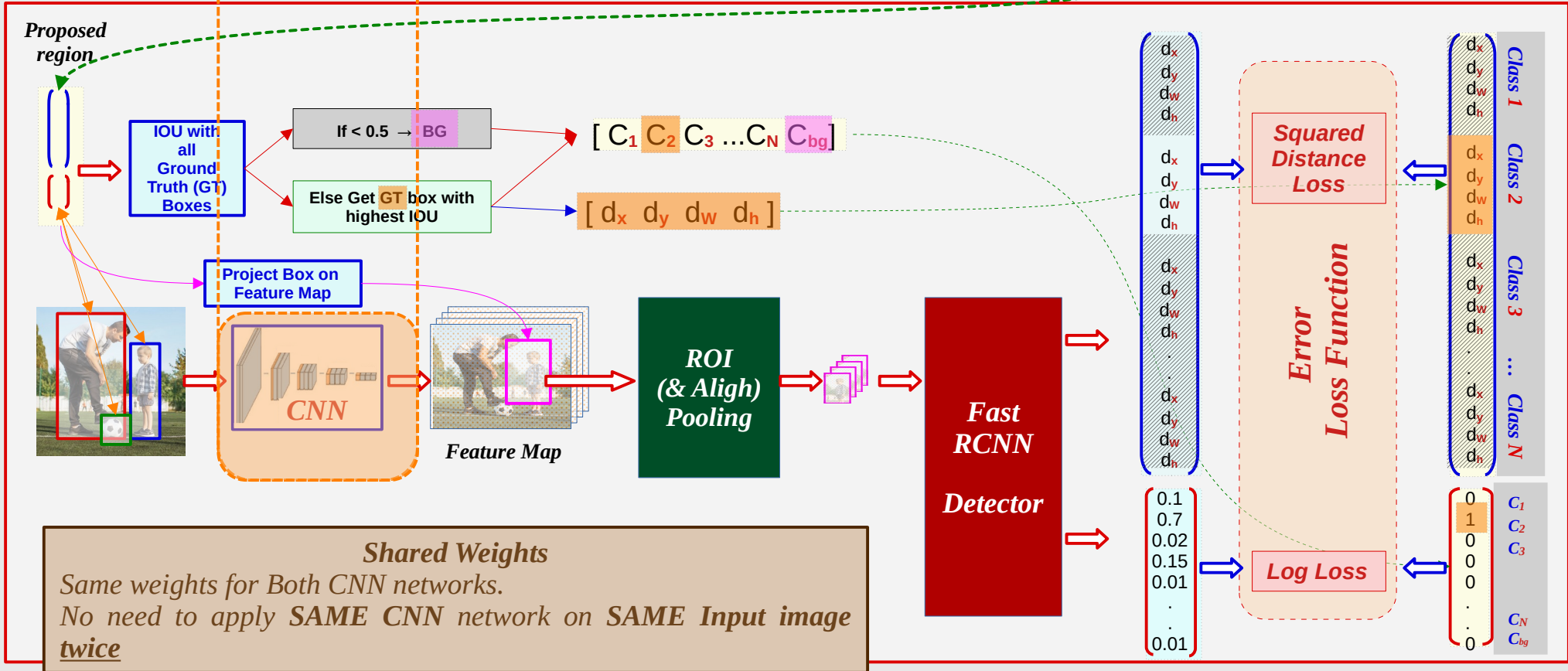
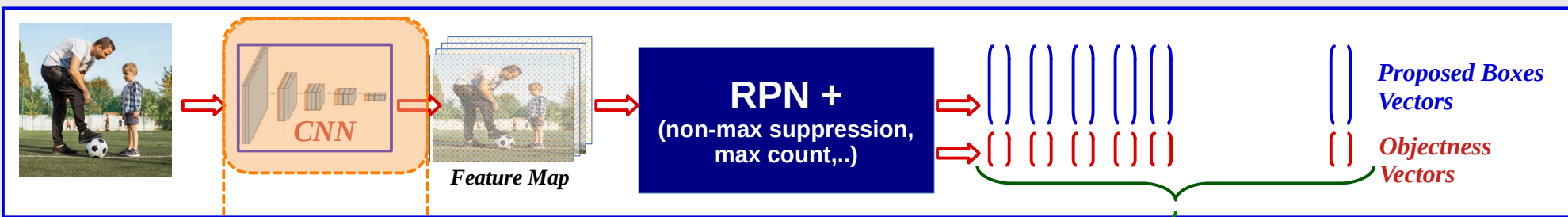


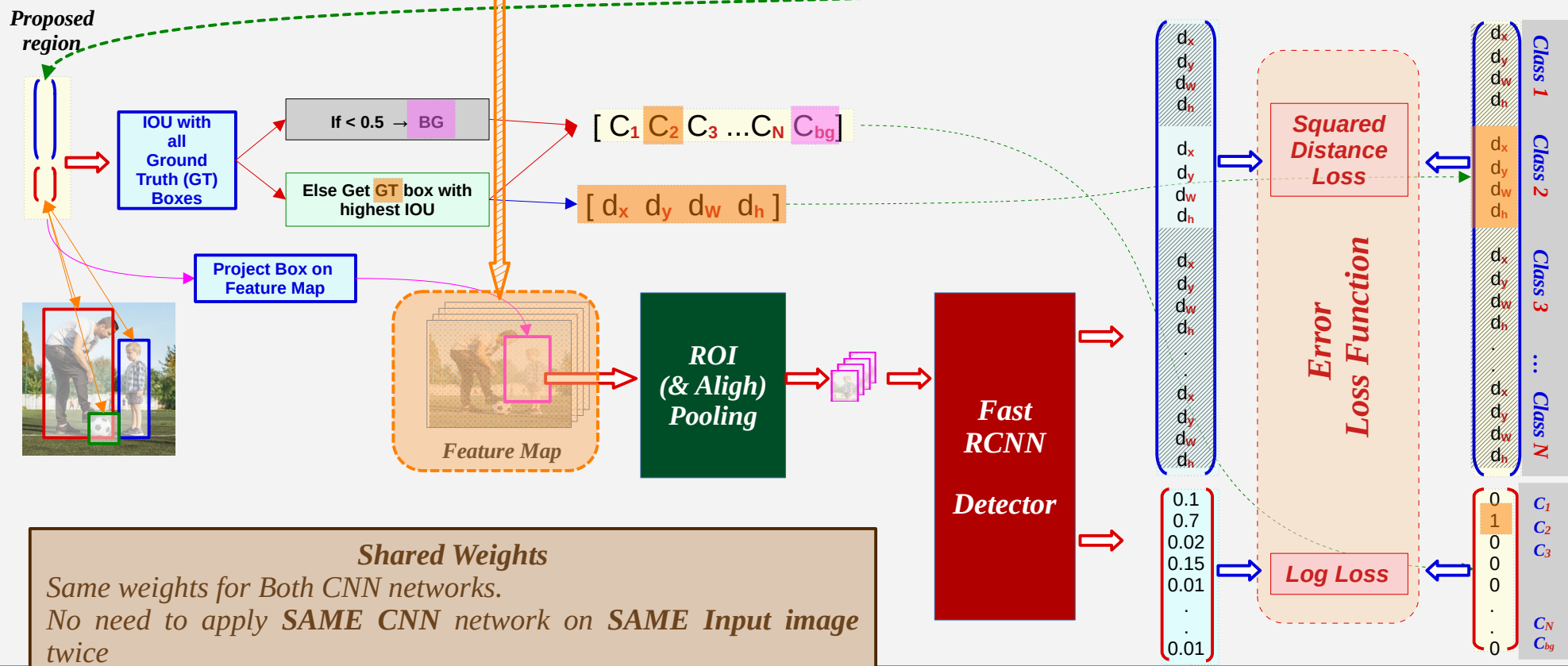
Proposed region



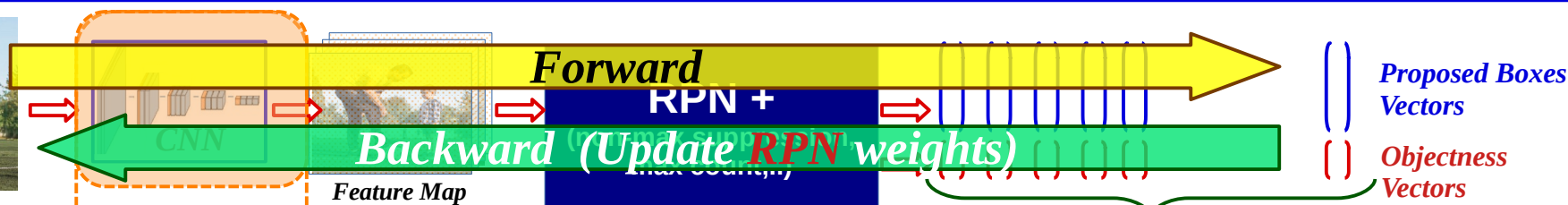




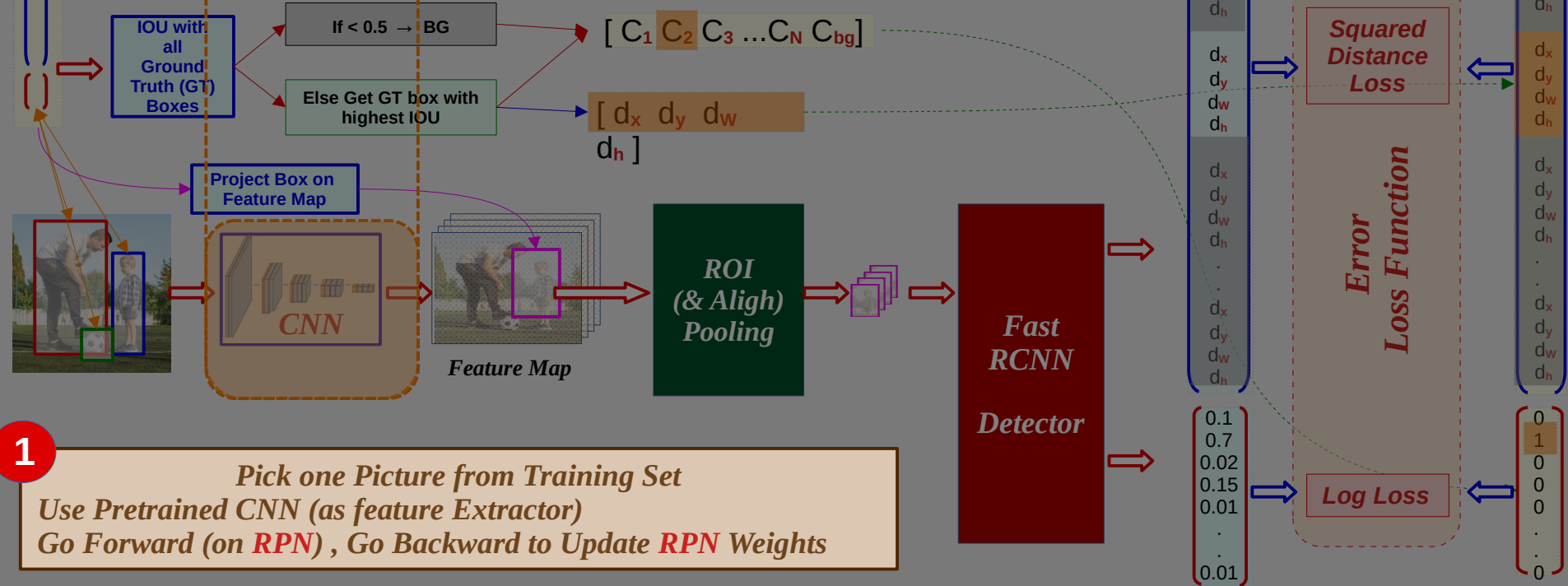


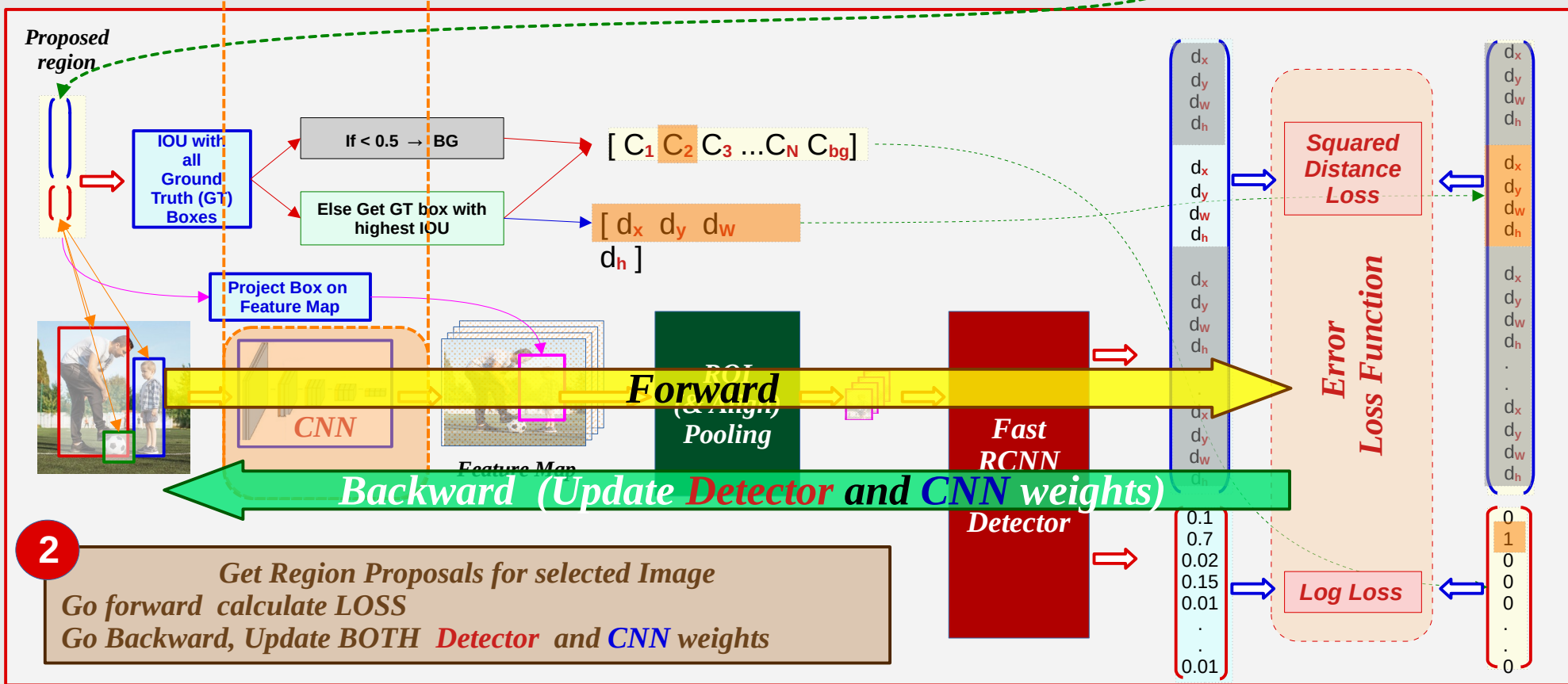
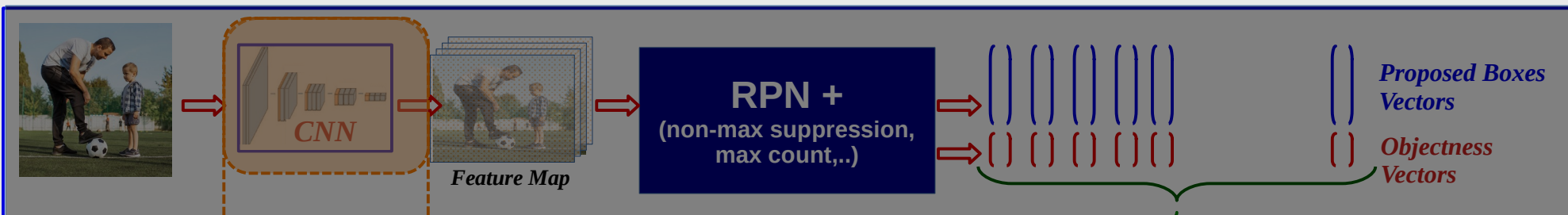


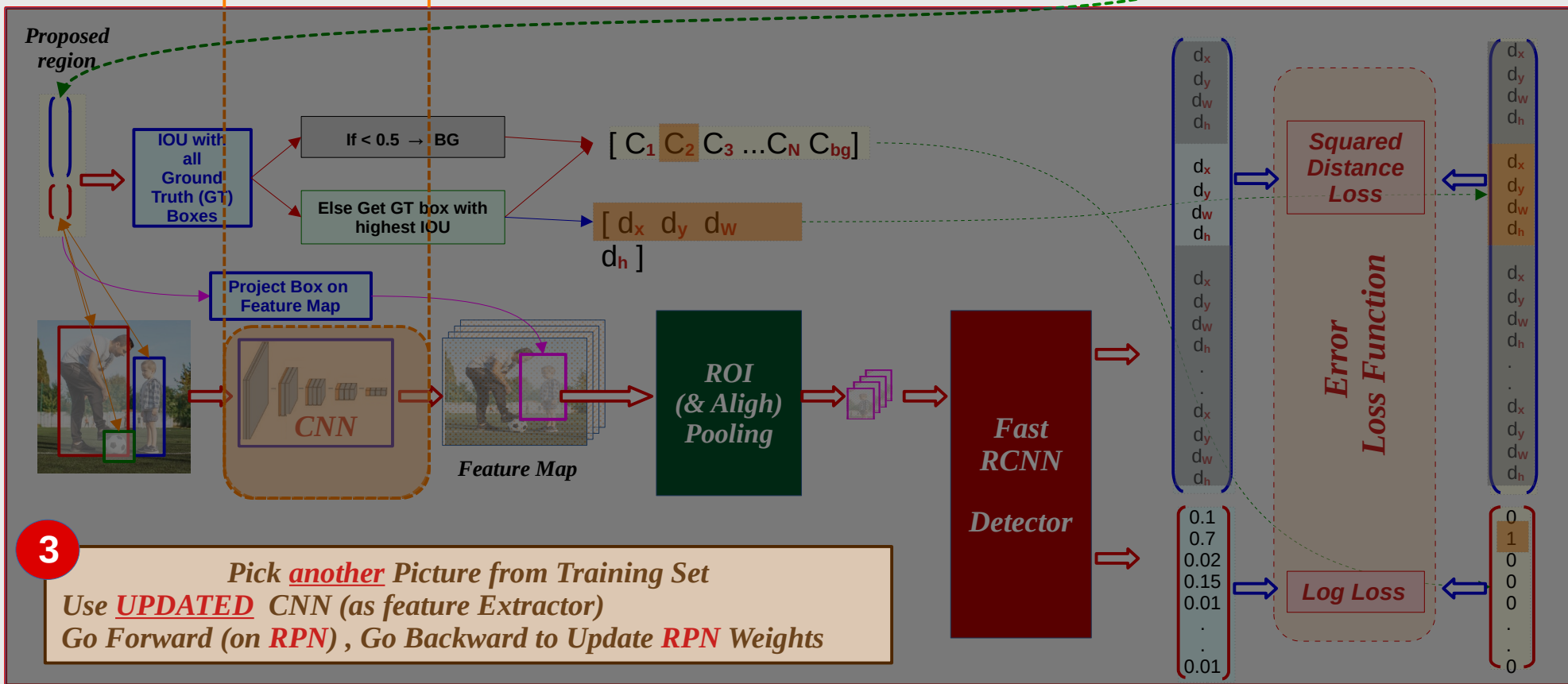
Alternate Training for RPN and Faster RCNN Detector

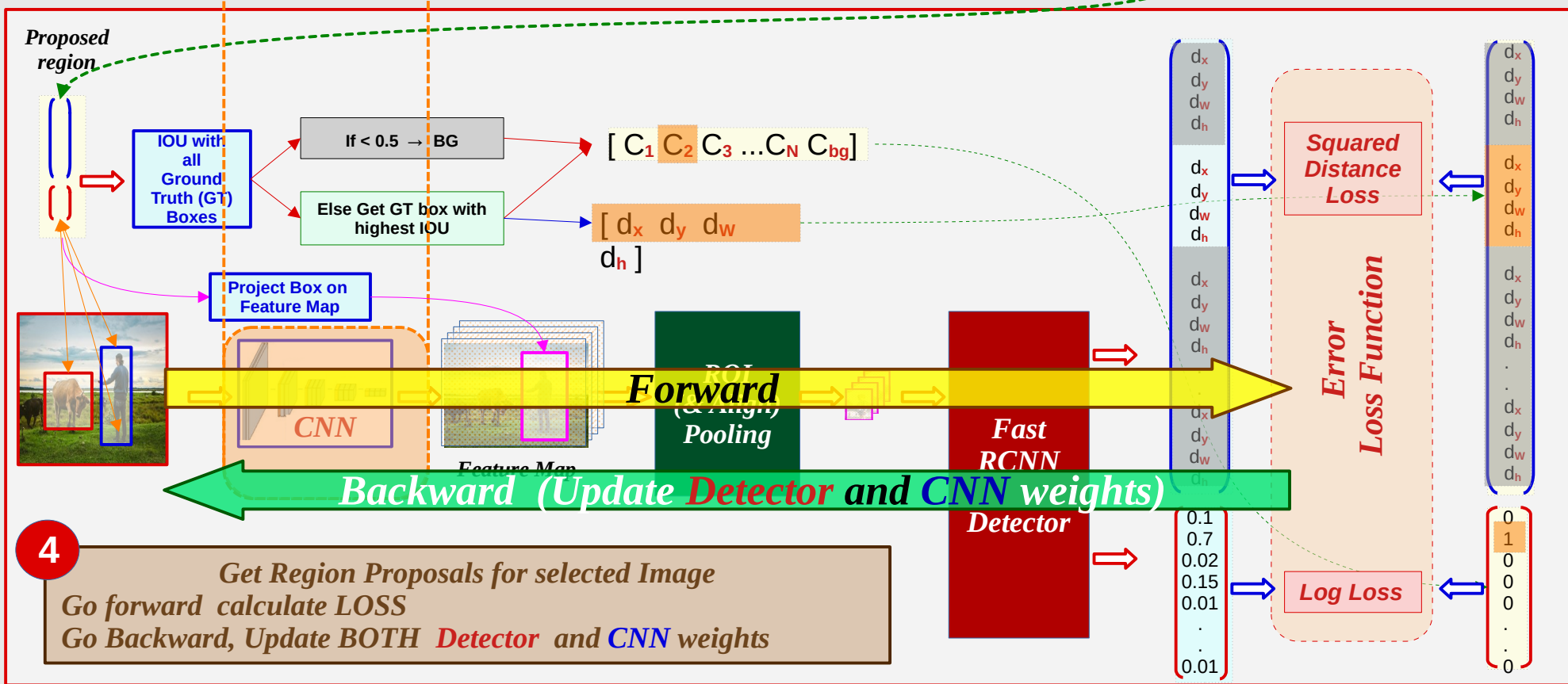
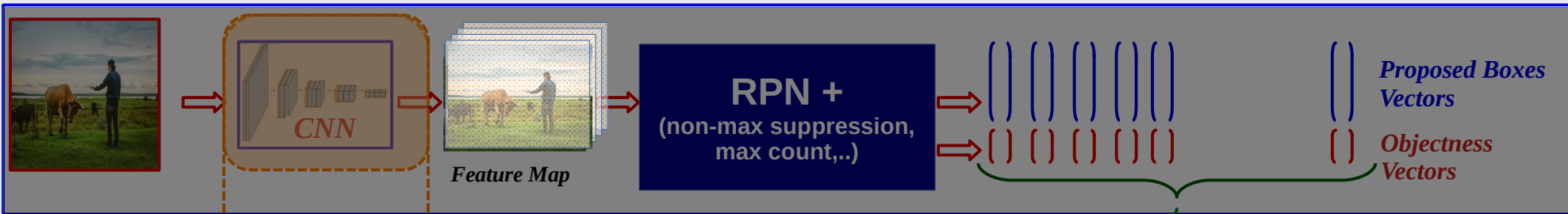


Proposed region











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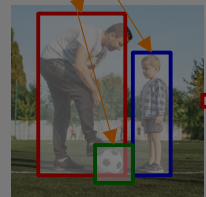
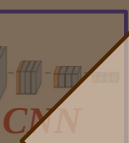
Proposed region

IOU with all Ground Truth (GT) Boxes

If $< 0.5 \rightarrow$ BG

Else Get GT box with highest IOU

Project Box on Feature Map



Feature Map

ROI (& Align) Pooling

Fast RCNN Detector

 d_x
 d_y
 d_w
 d_h
 d_x
 d_y
 d_w
 d_h
 $\cdot$
 d_x
 d_y
 d_w
 d_h

Squared Distance Loss

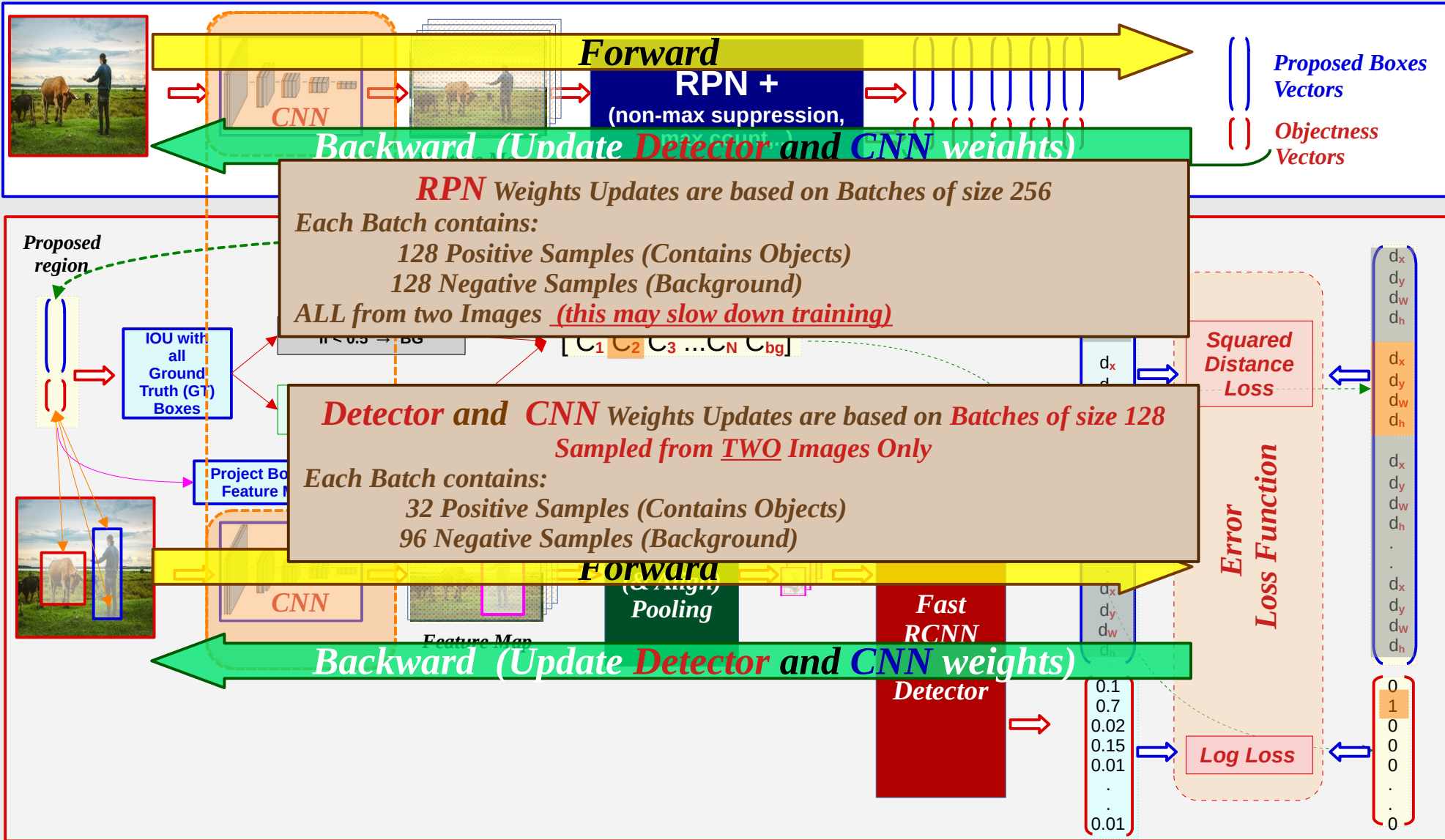
Error Loss Function

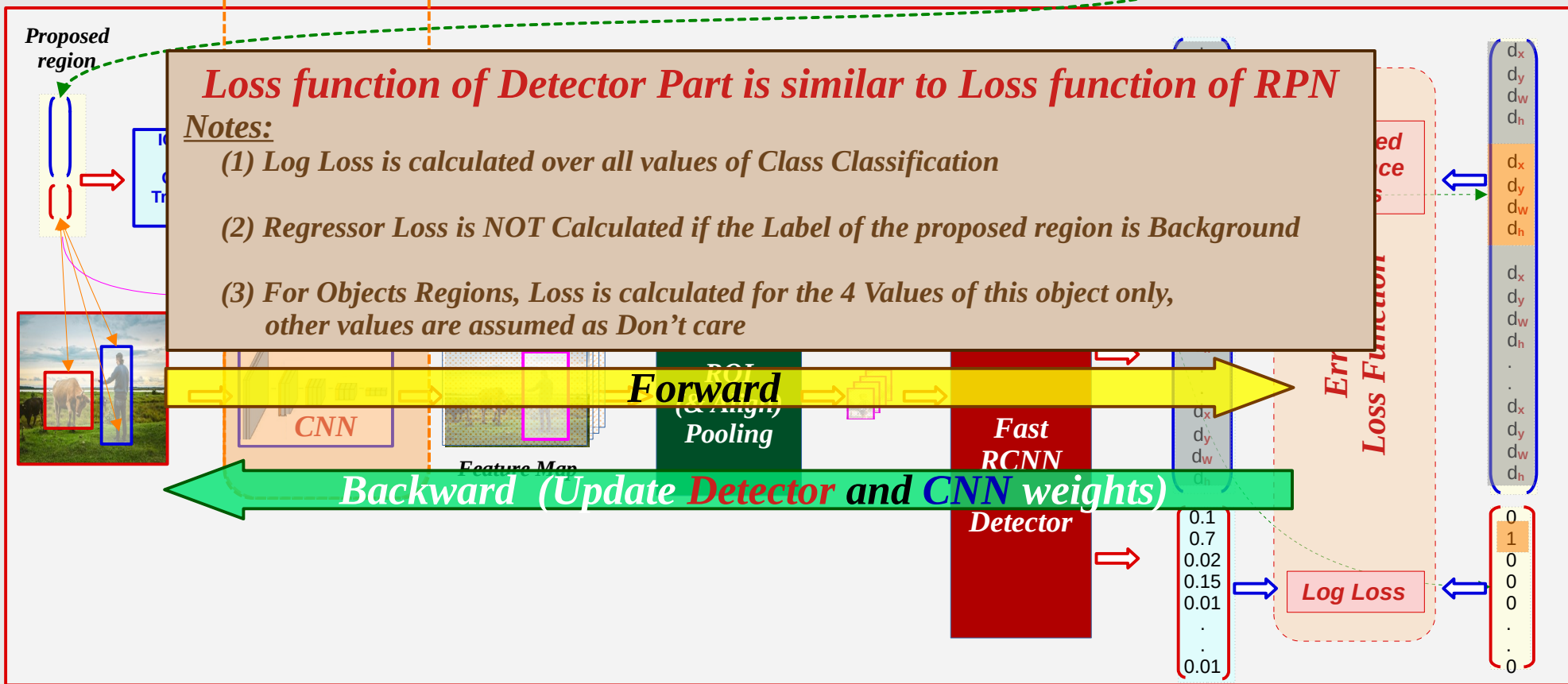
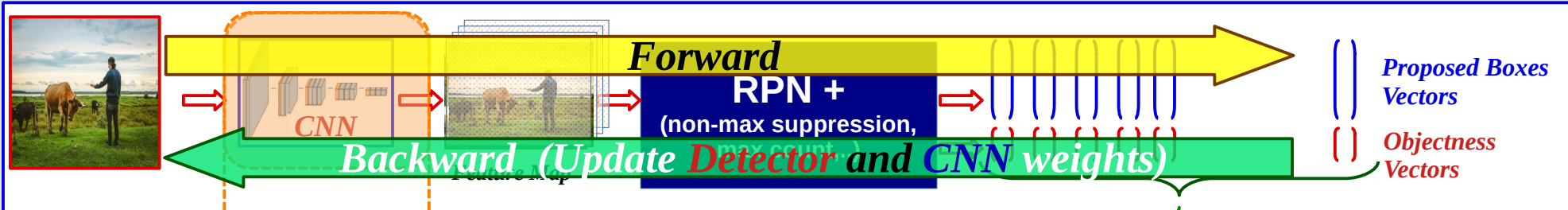
0.1
0.7
0.02
0.15
0.01
 $\cdot$
0.01

Log Loss

 d_x
 d_y
 d_w
 d_h
 d_x
 d_y
 d_w
 d_h
 $\cdot$
 d_x
 d_y
 d_w
 d_h 0
1
0
0
0
 $\cdot$
0

Continue Till Last Image In Training Samples



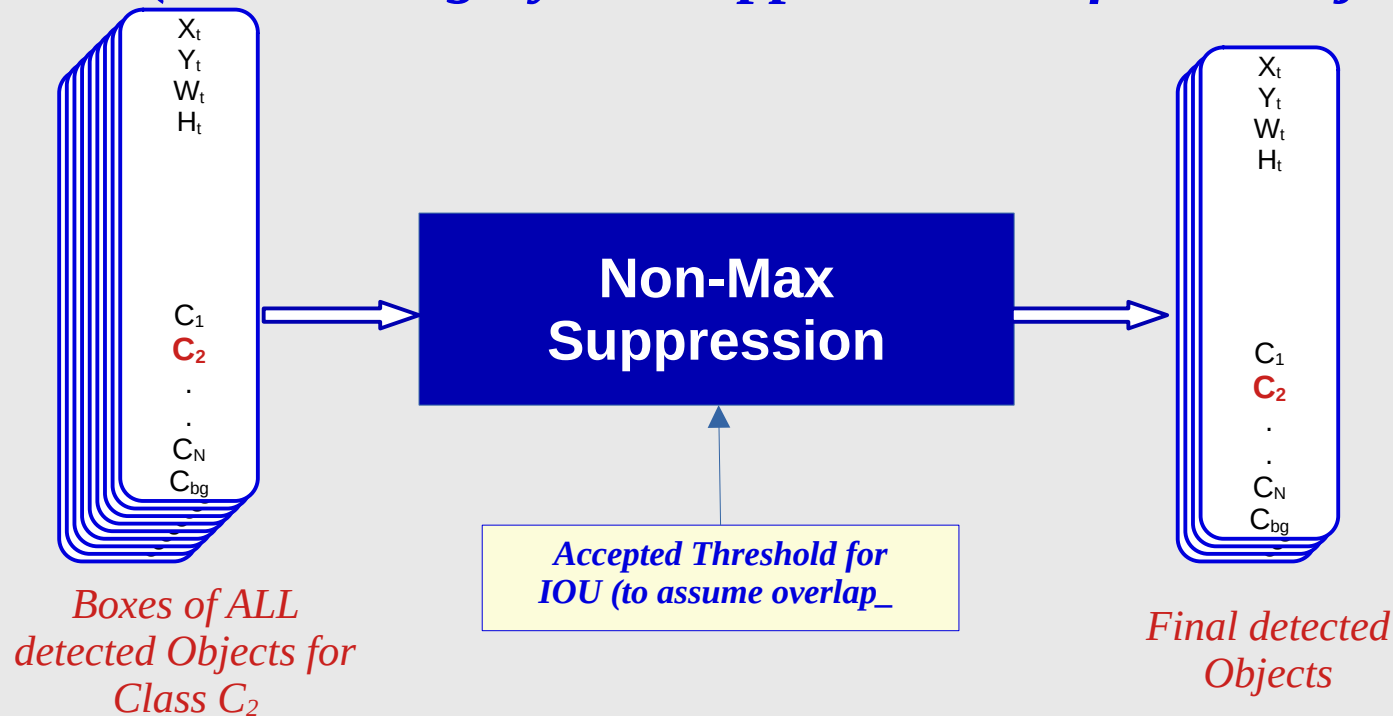


Final Detected Objects

using Detector and Non-Max Suppression

Filter Boxes of detected Objects (within same Object Class)

(Remove highly overlapped Boxes of same object)



Repeat for All Classes

Image Segmentation

Some images are captured from: [Fully Convolutional Networks for Semantic Segmentation paper \(2015\)](#)

Semantic Segmentation

One Color per Class (even if there are multiple instances of same class)



Original Image



Ground Truth



Segmented Image

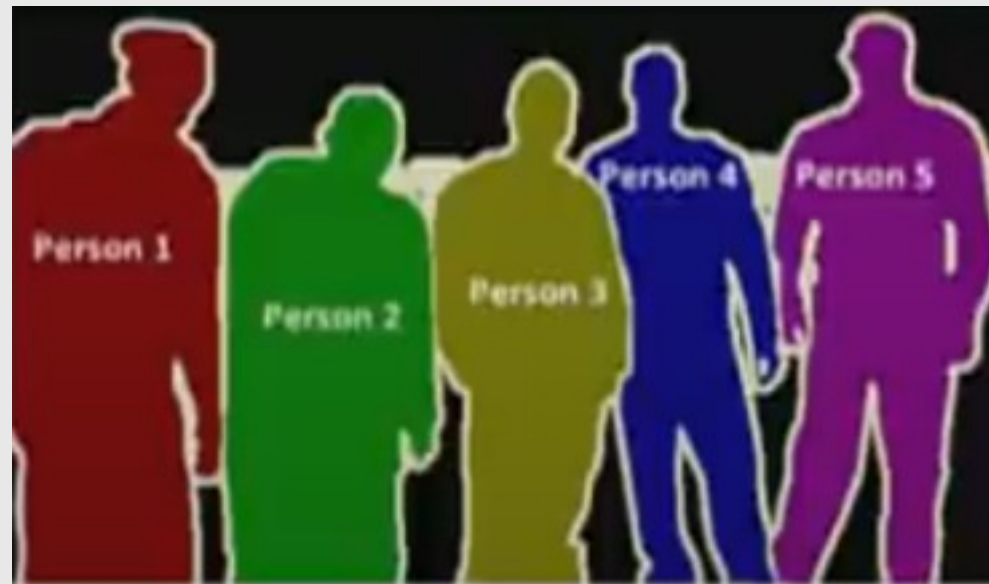
Semantic Segmentation:

Classify the object class of each pixel within an image. There is a label for each pixel

Semantic Segmentation vs Instance Segmentation

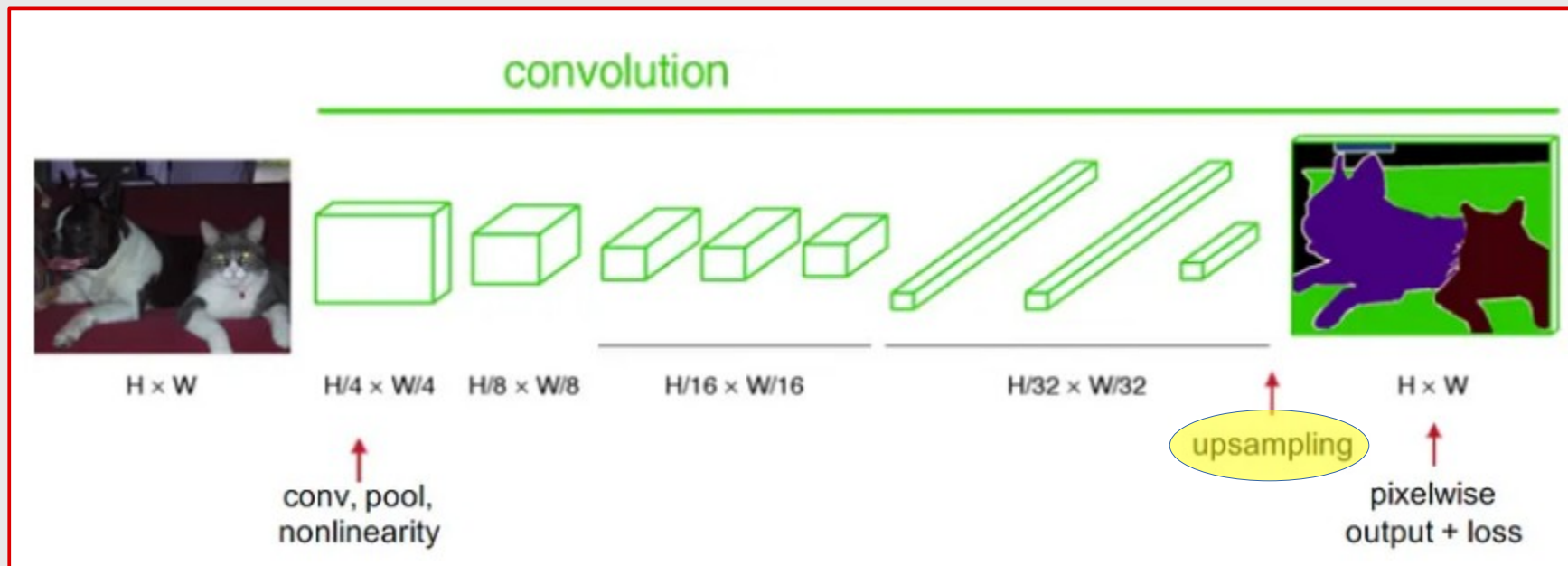


**Classification is based on:
Object Class**

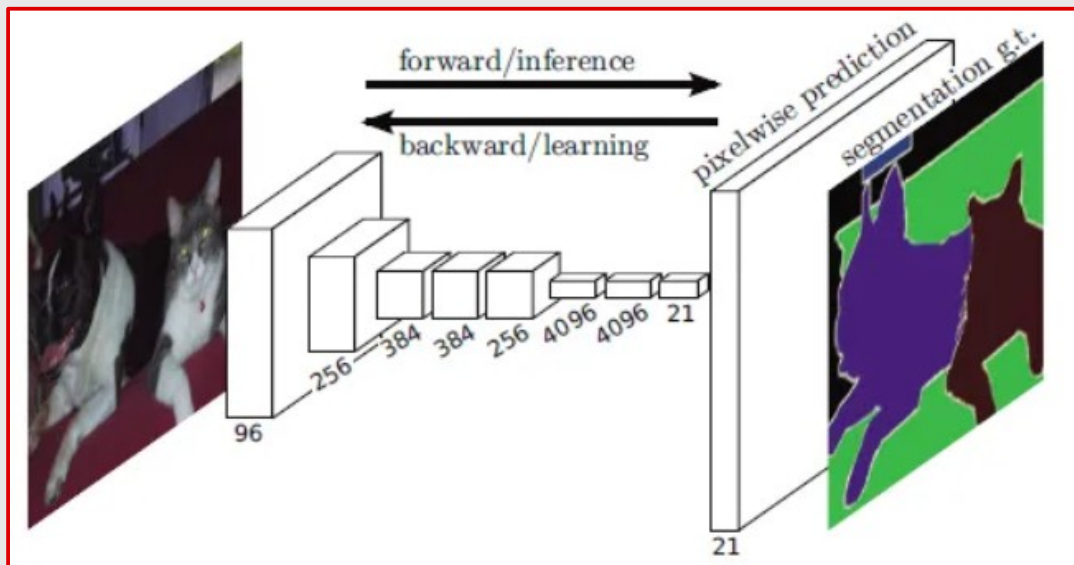
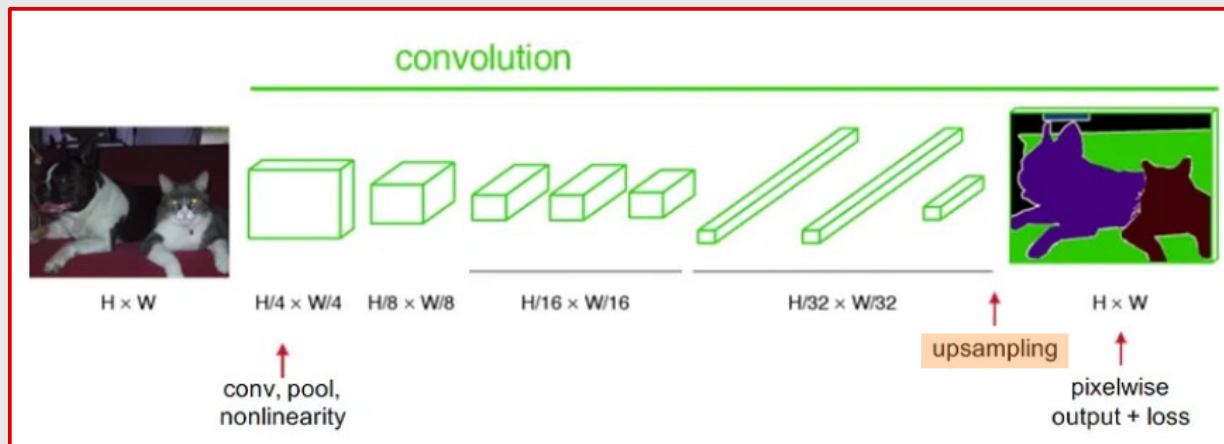


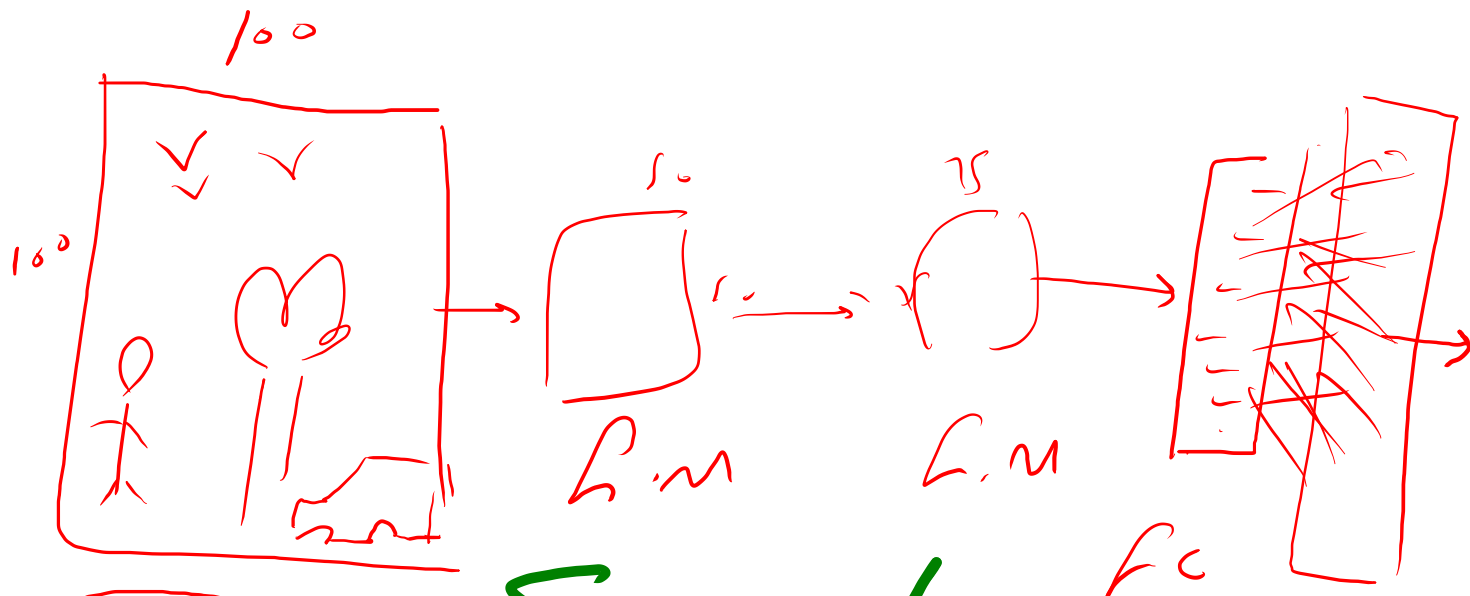
**Classification is based on:
Object Instance**

[1] Fully Convolutional Network (FCN)



[1] Fully Convolutional Network (FCN)





10,000

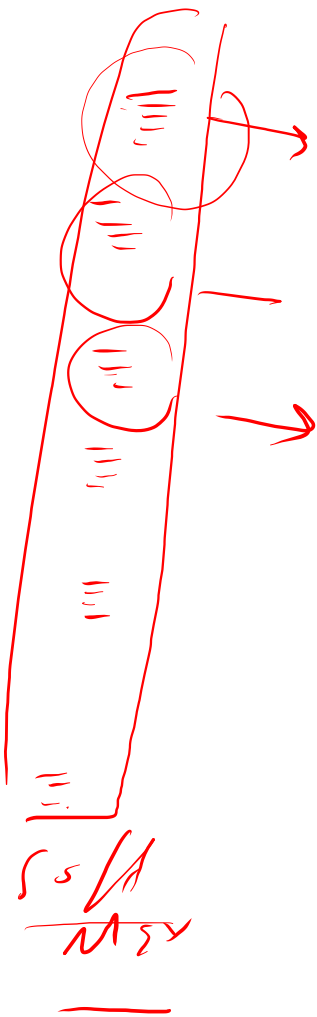
Semantic
Segmentation

pixels (class based)

assume C₀ C₁ ... C₇₉ 80 classes

Fc
up
sampling

10,000



[1] Fully Convolutional Network (FCN)

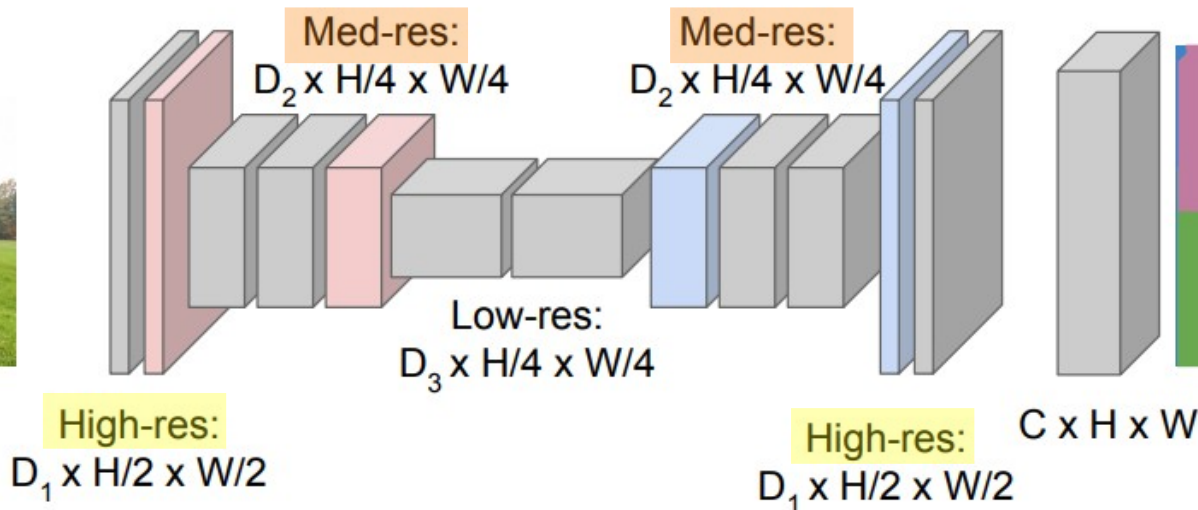
Downsampling:
Pooling, strided
convolution

Design network as a bunch of convolutional layers, with
downsampling and **upsampling** inside the network!

Upsampling:
???

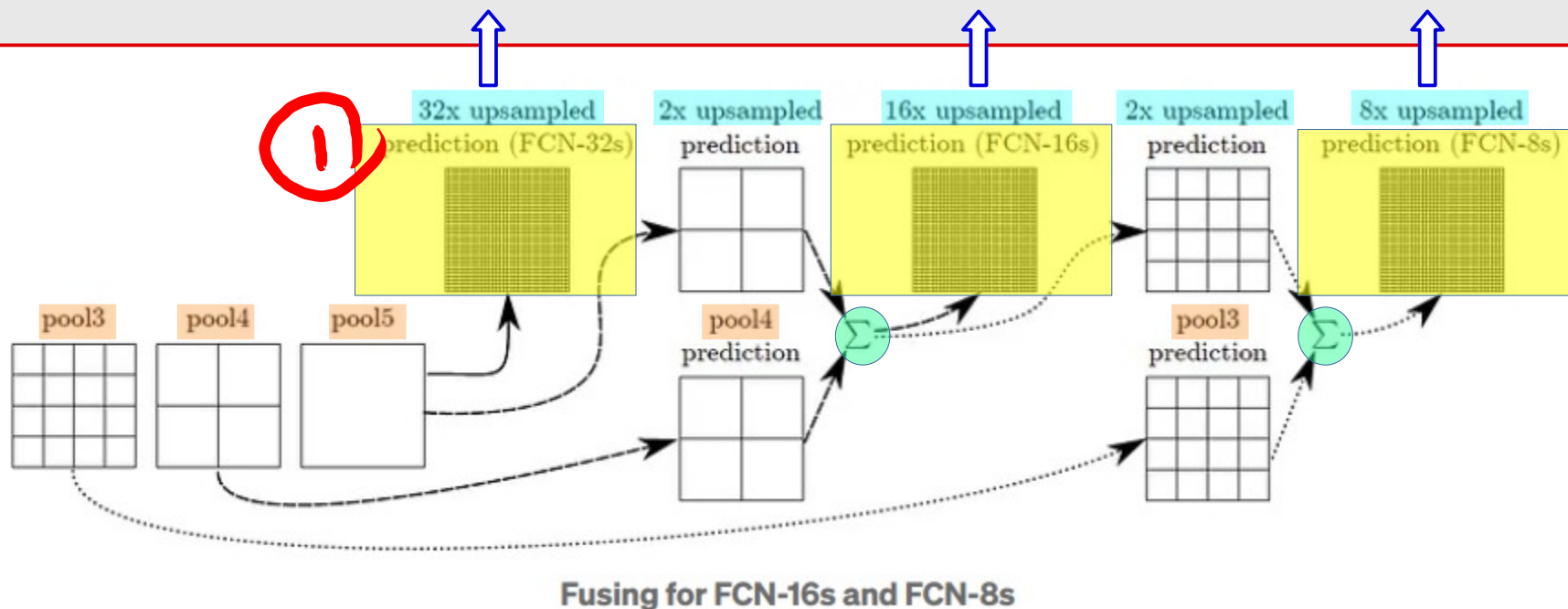
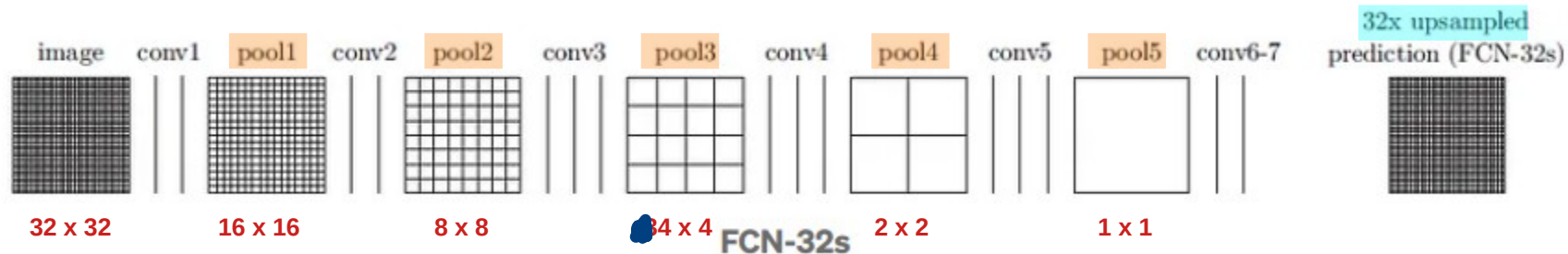


Input:
 $3 \times H \times W$

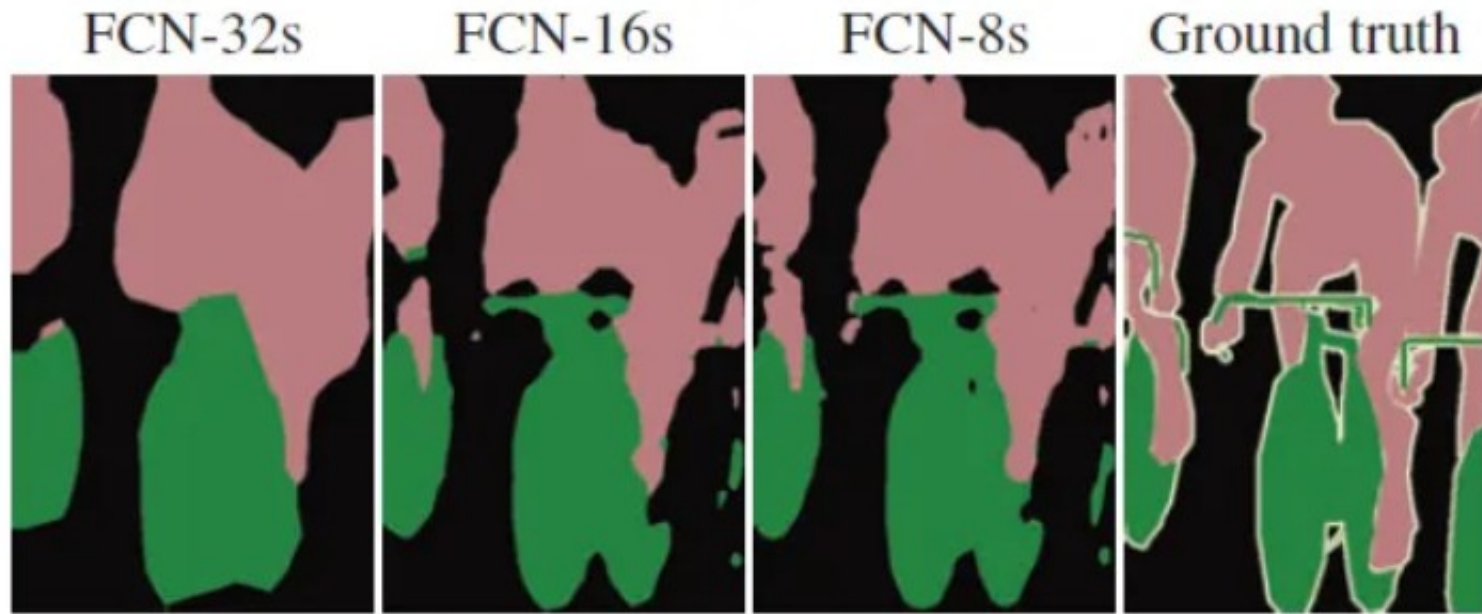


Predictions:
 $H \times W$

Fusing the Output (for Up-sampling)

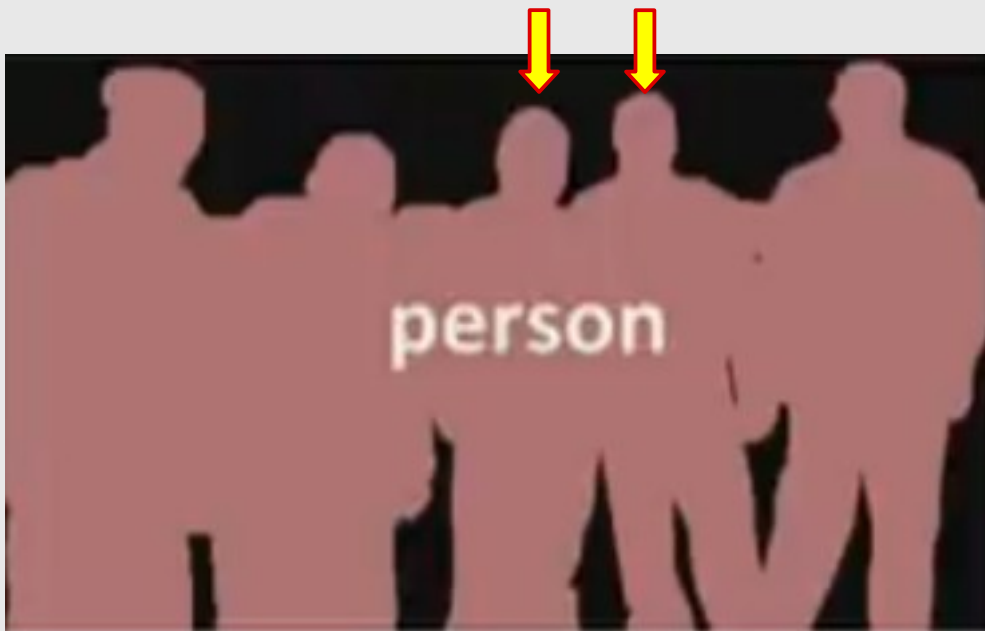


Fusing the Output



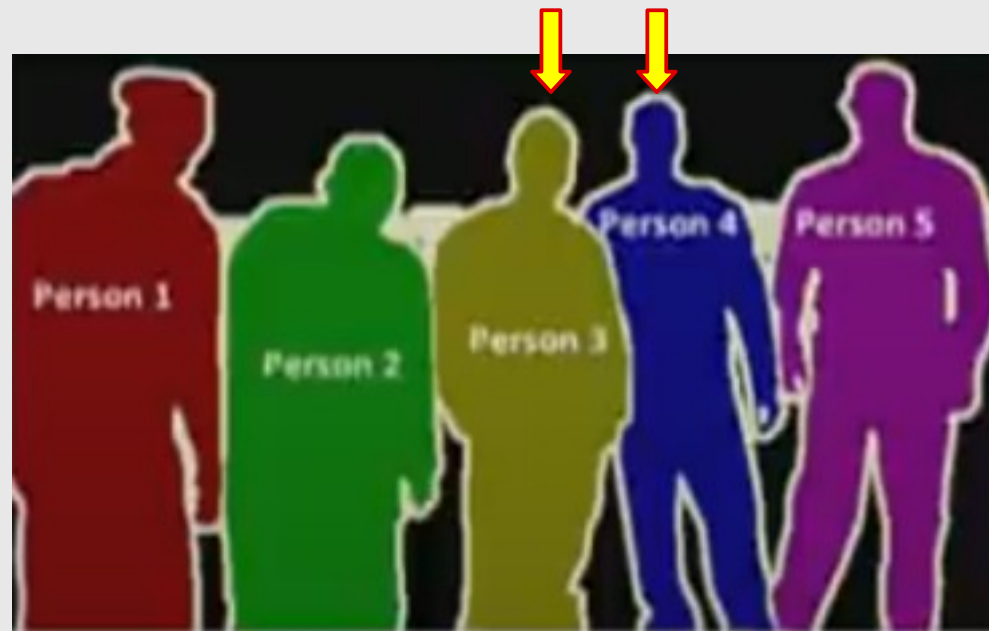
Comparison with different FCNs

Sematic Segmentation vs Instance Segmentation



**Classification is based on:
Object Class**

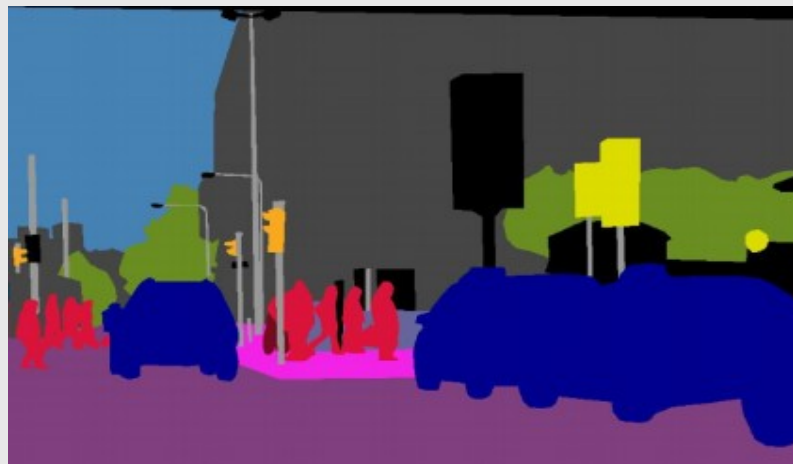
Overlapping Problem



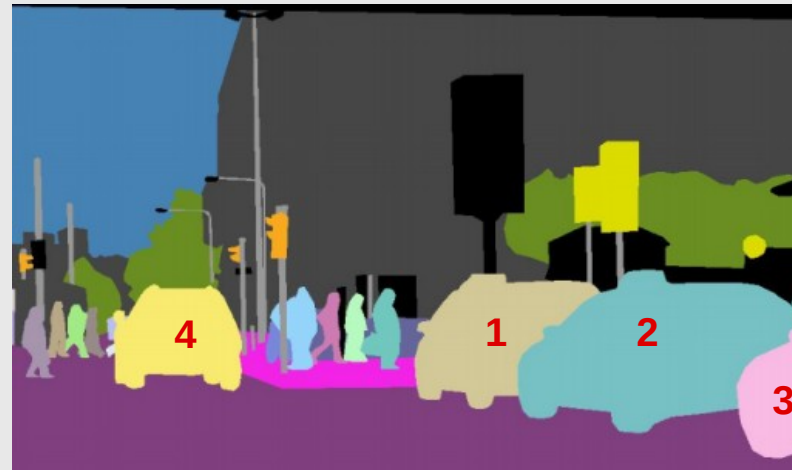
**Classification is based on:
Object Instance**

Solved Overlapping Problem

Sematic Segmentation vs Instance Segmentation



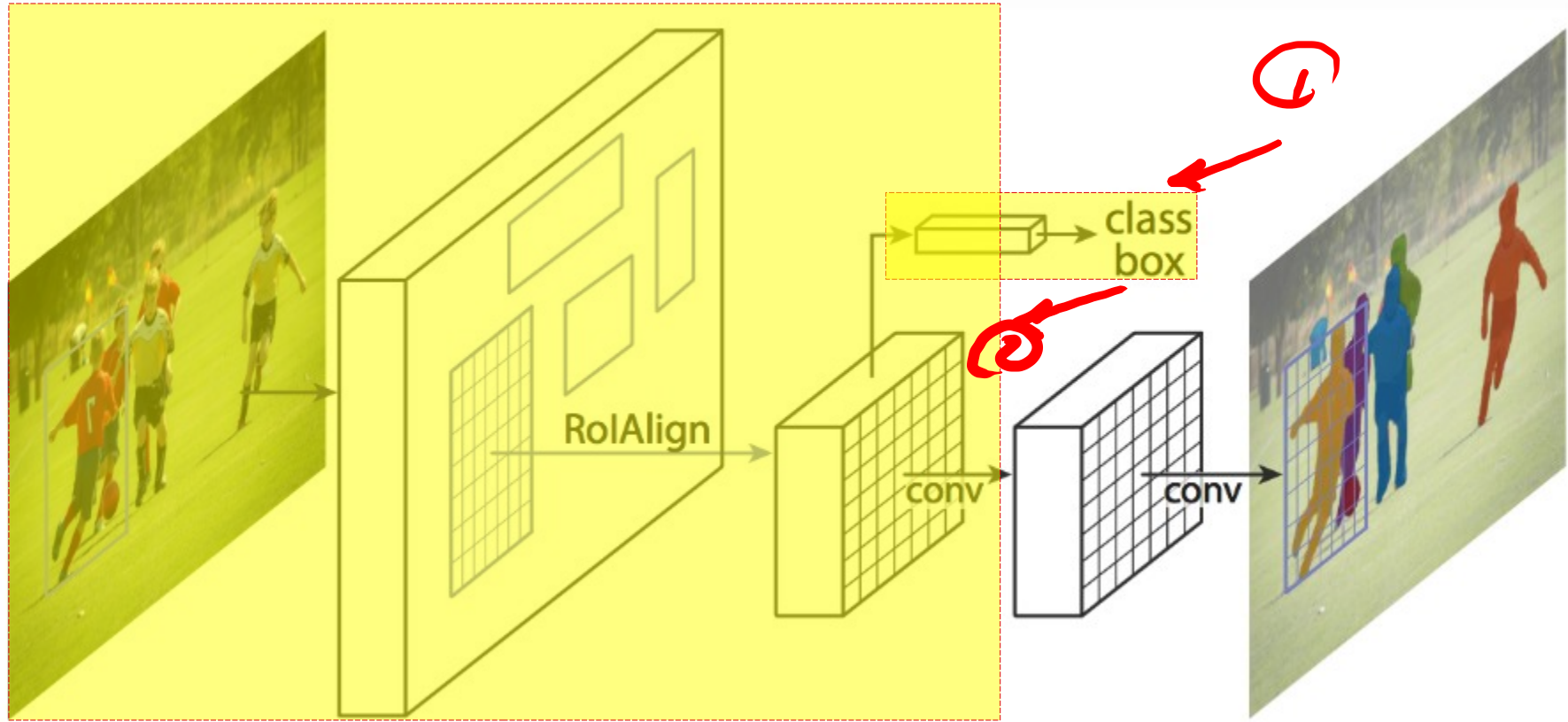
Classification is based on Object Class



Classification is based on Object Instance

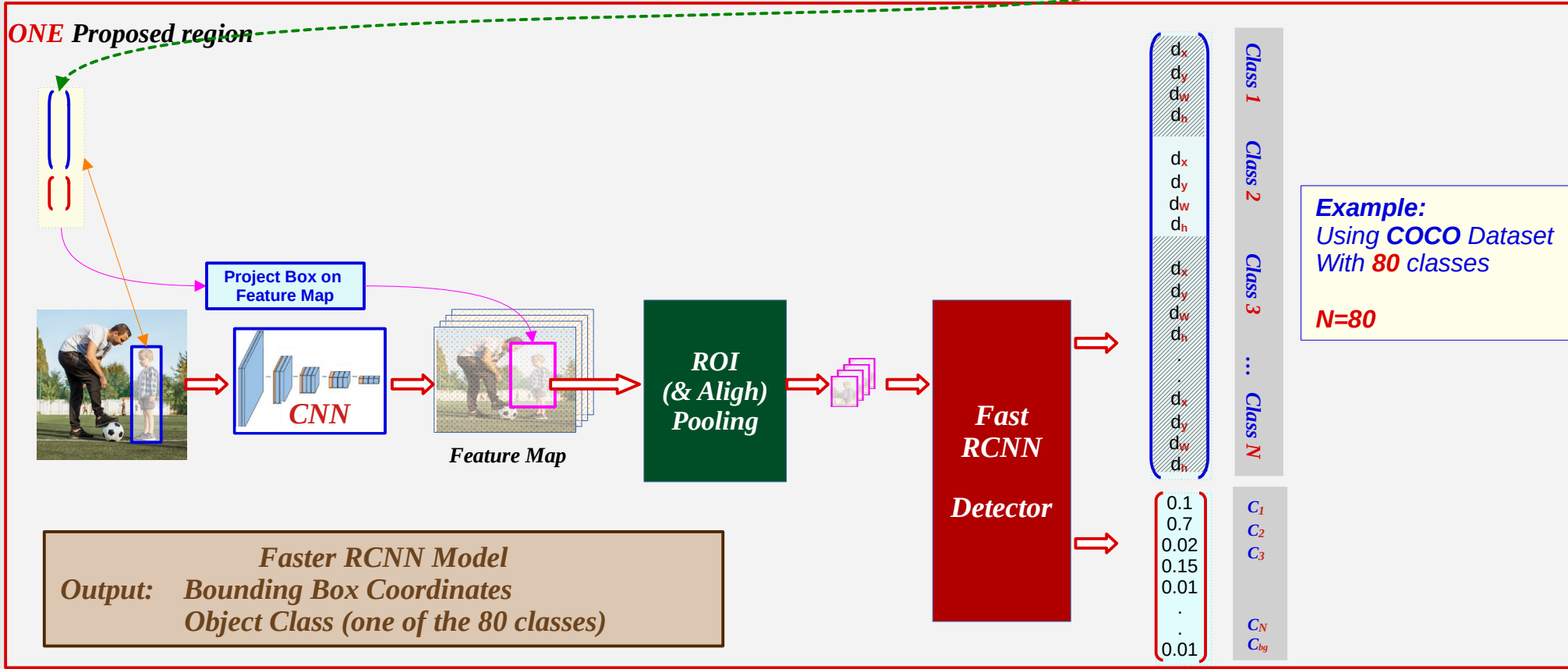
Mask RCNN

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Fast/Faster RCNN

Generated Mask

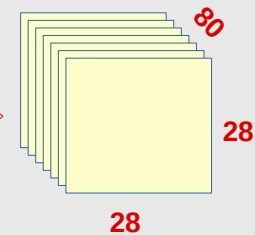


Output: *Faster RCNN Model*
Bounding Box Coordinates
Object Class (one of the 80 classes)

Mask RCNN (Using FPN Backbone)

34

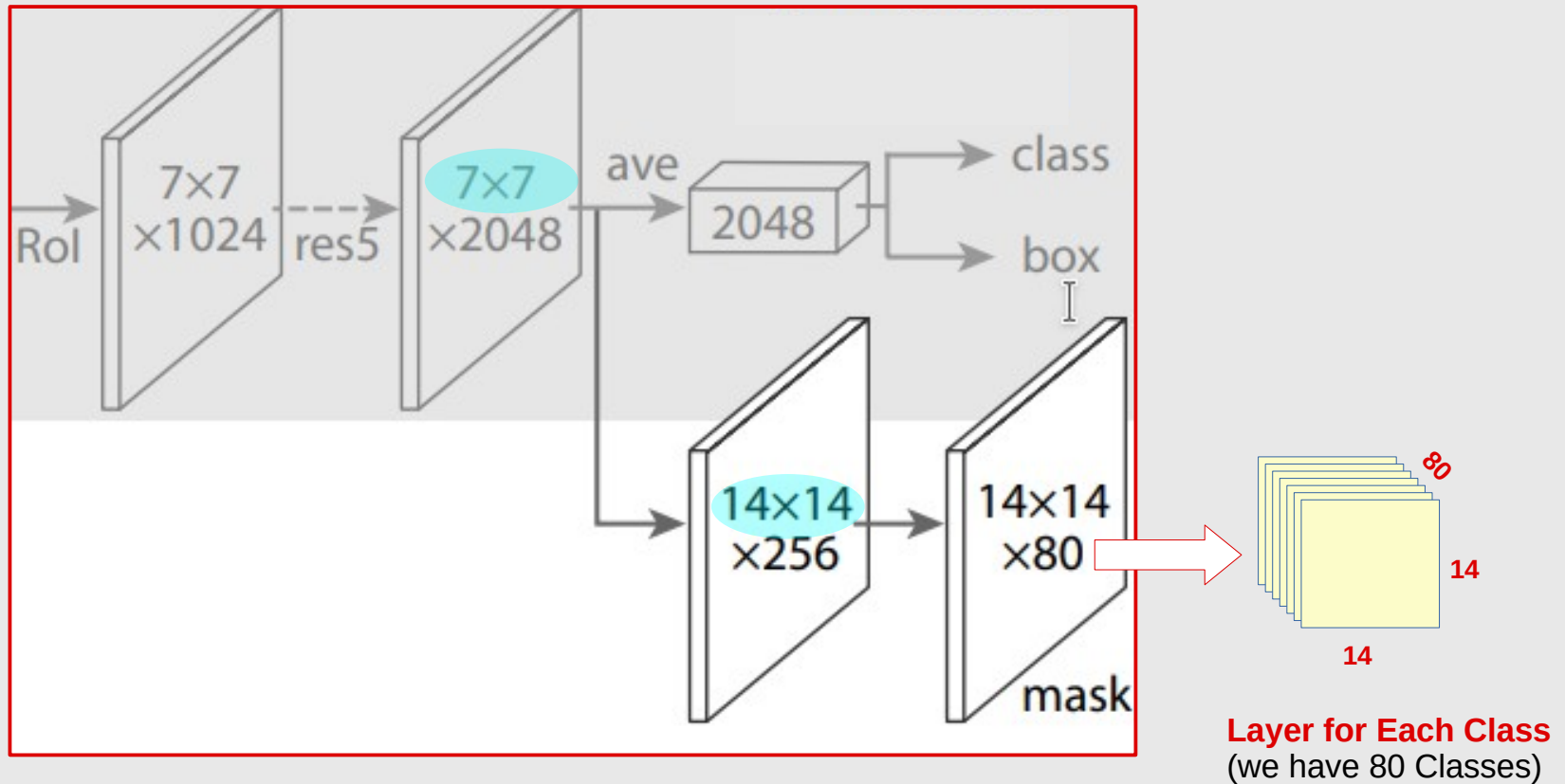
*RCNN
Detect*



Layer for Each Class
(we have 80 Classes)

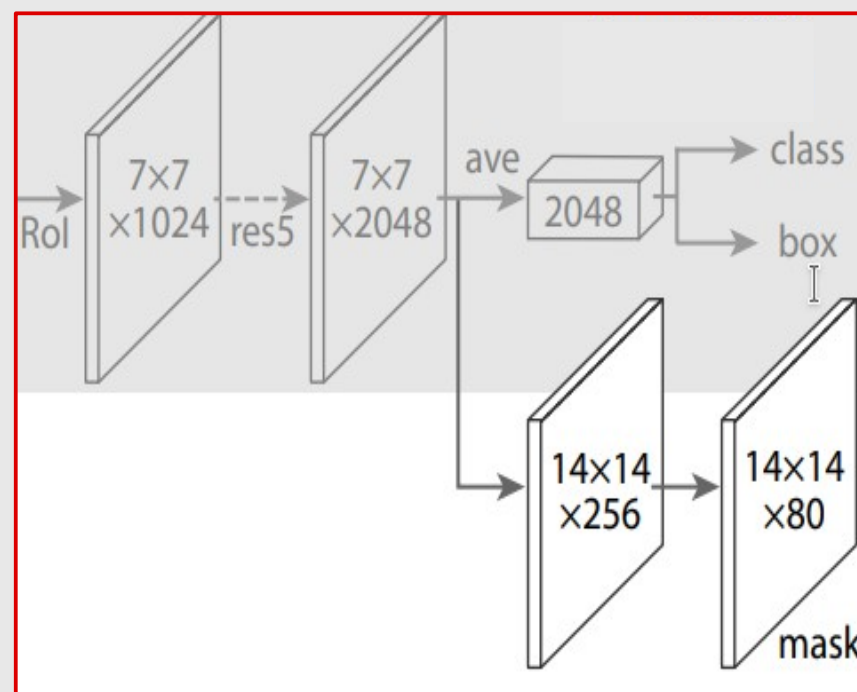
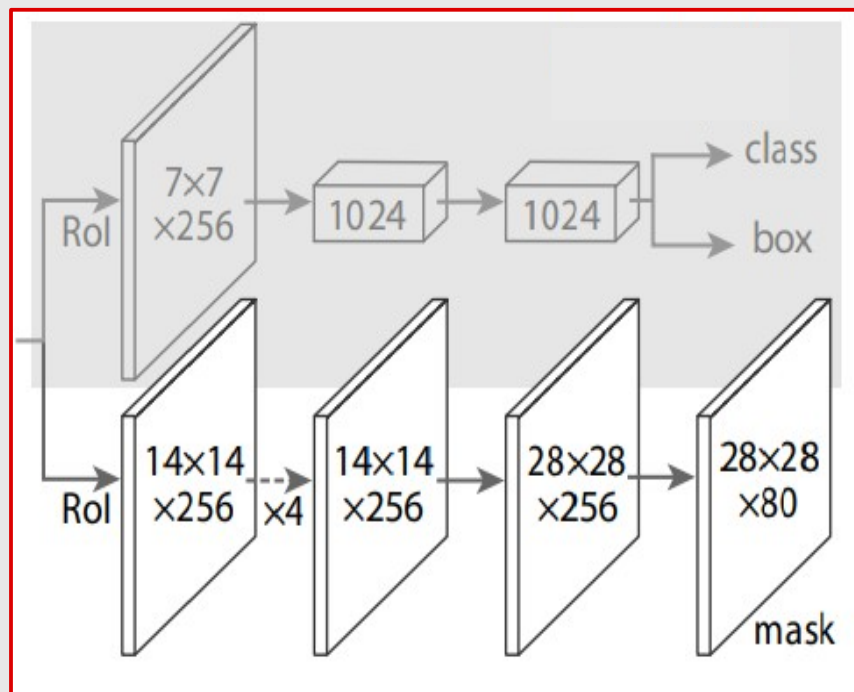
Mask RCNN (Using ResNet Backbone)

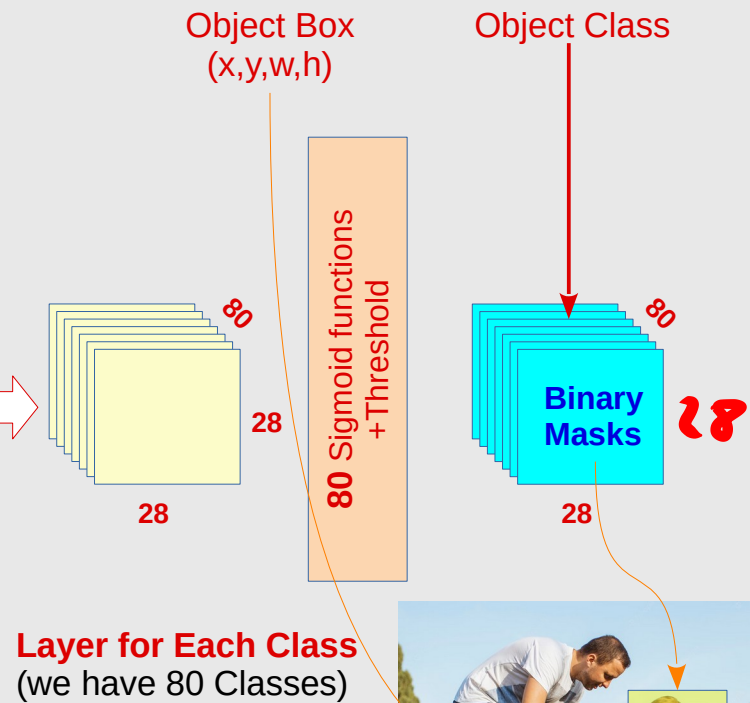
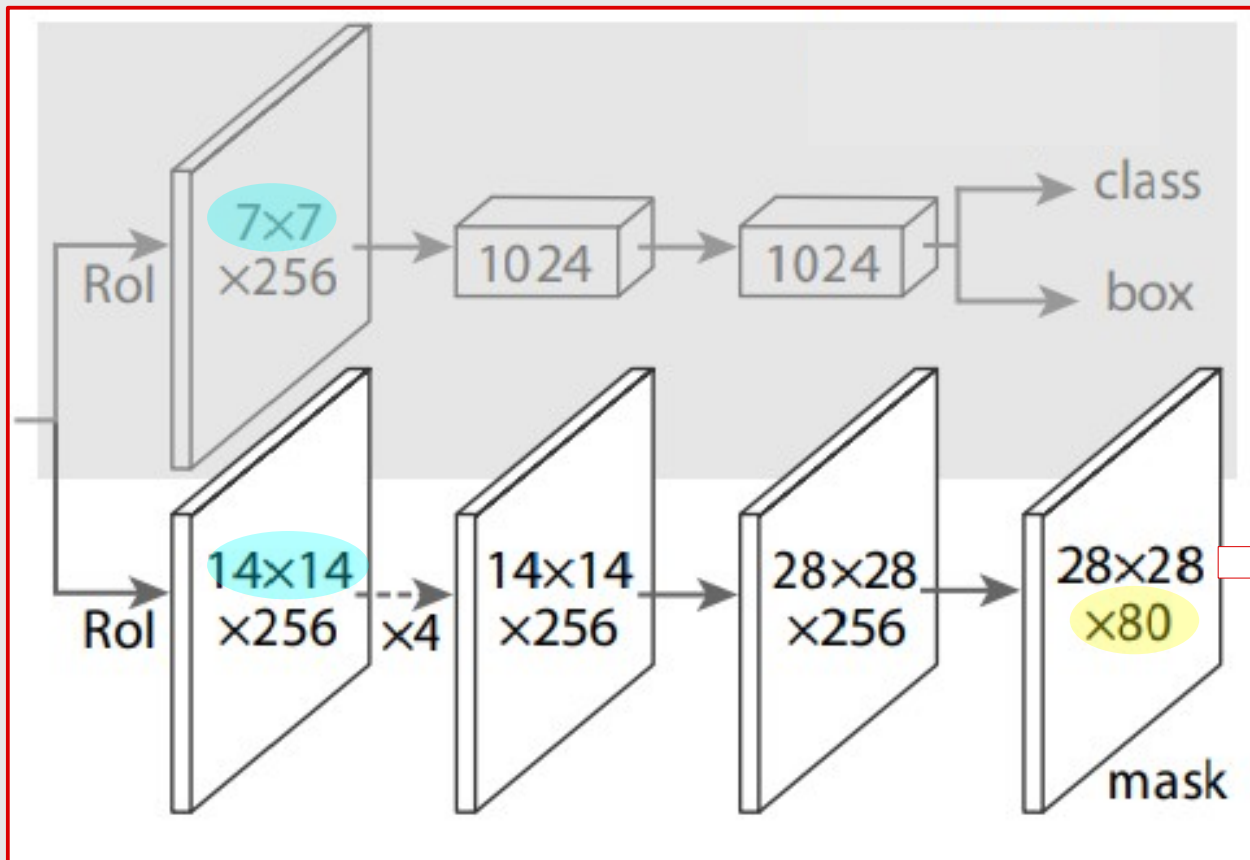
35



Mask RCNN

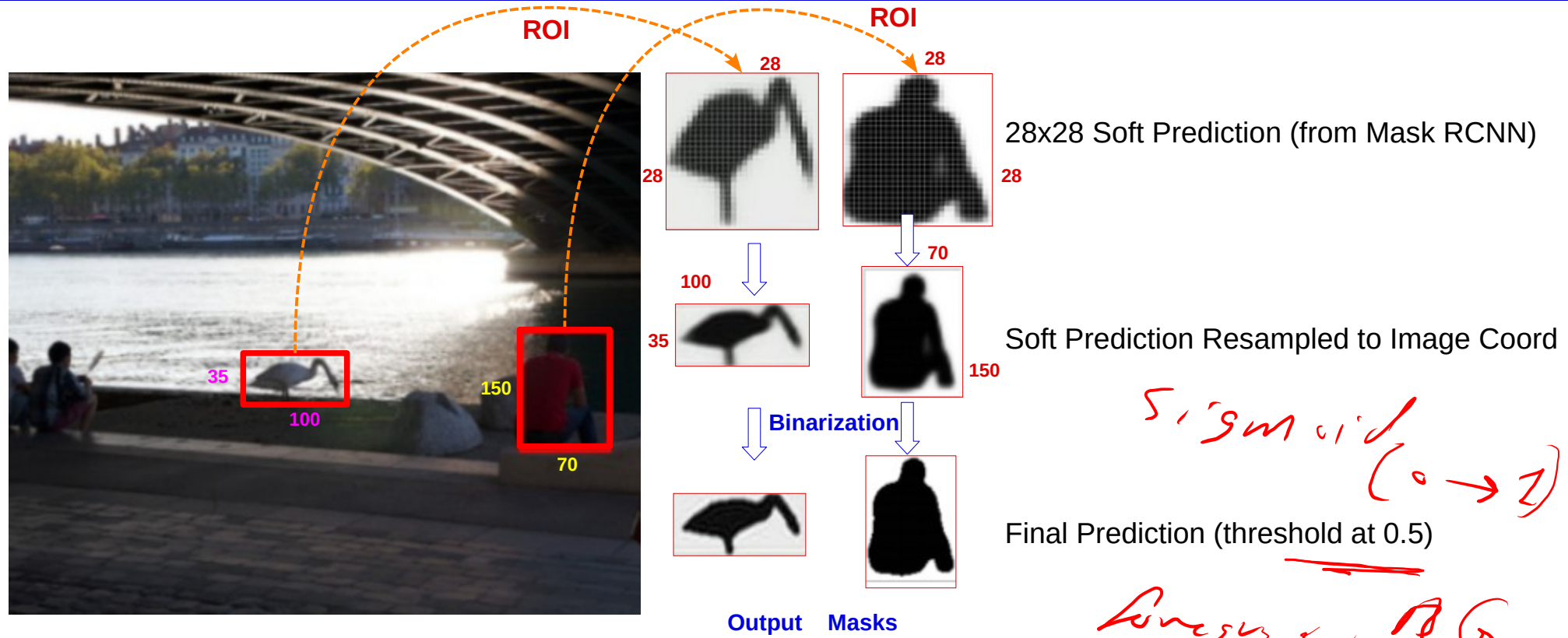
(using different CNN Backbones)





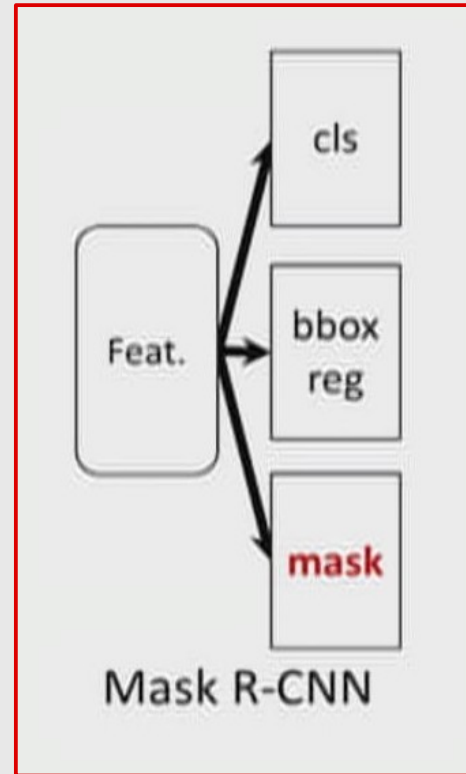
Mask RCNN (Final Prediction Mask)

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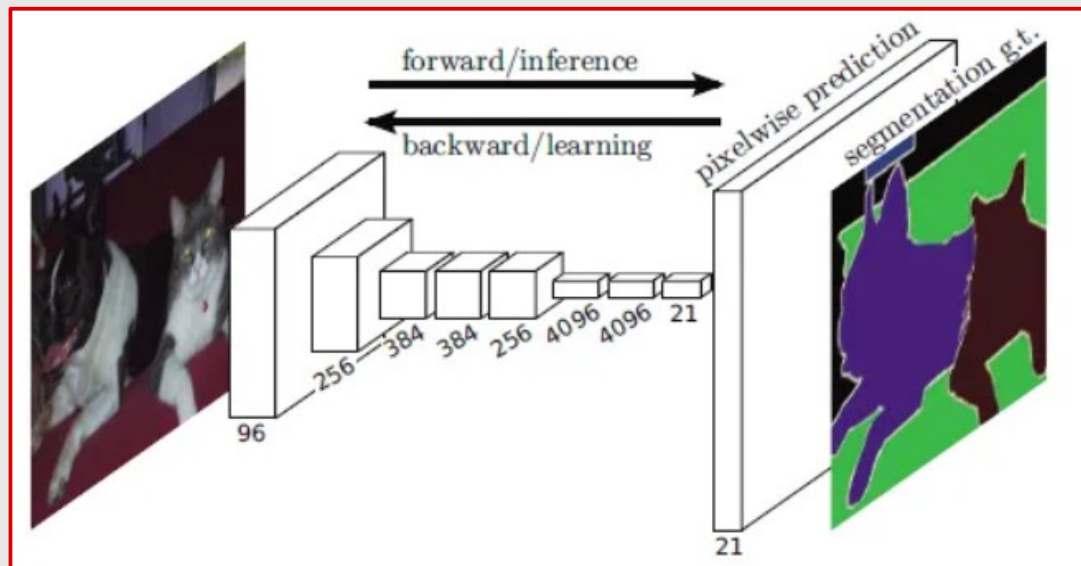
Mask RCNN (Loss Function)

$$L = L_{cls} + L_{box} + L_{mask}$$



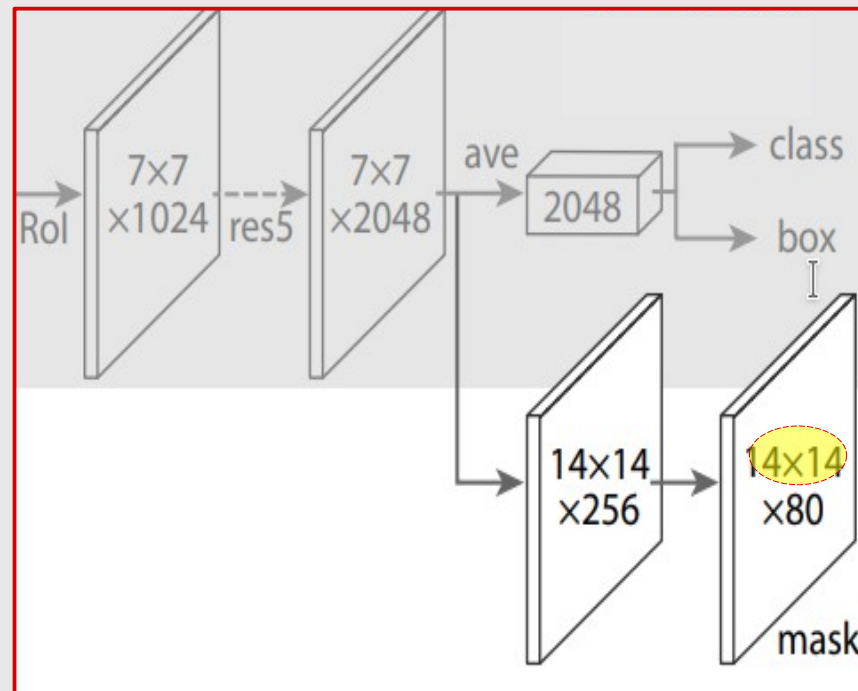
Multinomial vs. Independent Masks

FCN



80 Classes
 One Masks (for All classes)
 Use Softmax (among all classes)

Mask - RCNN



80 Classes
 80 Binary Masks *← = = ≠ 1 classes*
 Use Sigmoid (Binary Classification)

Multinomial vs. Independent Masks

	AP	AP ₅₀	AP ₇₅
<i>softmax</i>	24.8	44.1	25.1
<i>sigmoid</i>	30.3	51.2	31.5
	+5.5	+7.1	+6.4

FCN

Mask - RCNN

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